

INFORME DE ACTIVIDADES EN EL MARCO DEL PROGRAMA:
ALIANZA CIENTÍFICA CON ENFOQUE
COMUNITARIO PARA MITIGAR BRECHAS DE
ATENCIÓN Y MANEJO DE TRASTORNOS
MENTALES RELACIONADOS CON IMPULSIVIDAD
EN COLOMBIA

Nombre: Julián David Pastrana Cortés

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GRUPO DE INVESTIGACIÓN EN AUTOMÁTICA

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1. Información general del contrato

Rol	Contratista - Estudiante de Doctorado
Contrato de servicios No.	5533 de 2026
Objeto del contrato	Prestación de servicios profesionales para el Desarrollo de metodología de gamificación para entrenamiento de niños con trastornos de impulsividad ALIANZA CIENTÍFICA CON ENFOQUE COMUNITARIO PARA MITIGAR BRECHAS DE ATENCIÓN Y MANEJO DE TRASTORNOS MENTALES RELACIONADOS CON IMPULSIVIDAD EN COLOMBIA - ACEMATE MINCIENCIAS CONTRATO 790-2023
Período del informe	19 de enero de 2026 al 31 de agosto de 2026

1.1. Descripción general de la vinculación

Como resultado de la vinculación se contribuye al alcance de el objetivo 3 del programa de investigación: ALIANZA CIENTÍFICA CON ENFOQUE COMUNITARIO PARA MITIGAR BRECHAS DE ATENCIÓN Y MANEJO DE TRASTORNOS MENTALES RELACIONADOS CON IMPULSIVIDAD EN COLOMBIA

1.2. Objetivo general

Desarrollar una metodología de gamificación para entrenamiento de niños con trastornos de impulsividad ALIANZA CIENTÍFICA CON ENFOQUE COMUNITARIO PARA MITIGAR BRECHAS DE ATENCIÓN Y MANEJO DE TRASTORNOS MENTALES RELACIONADOS CON IMPULSIVIDAD EN COLOMBIA - ACEMATE MINCIENCIAS CONTRATO 790-2023.

2. Metodología

La metodología propuesta para el desarrollo del proyecto se estructura en cuatro actividades principales, distribuidas a lo largo de siete meses, con entregables específicos para cada una:

2.1. Actividad 1: Planteamiento de la metodología, técnicas y modelos a implementar (Meses 1-2)

En esta etapa inicial se realizará la conceptualización y diseño de la metodología de trabajo. Se definirán las técnicas de análisis y los modelos computacionales que se implementarán para el procesamiento de señales EEG. Esta actividad incluye la revisión de fundamentos teóricos, selección de herramientas y establecimiento del marco de trabajo.

Entregable: Informe técnico que documenta la metodología propuesta, las técnicas seleccionadas y los modelos a desarrollar.

2.2. Actividad 2: Revisión técnica del estado del arte (Meses 3-4)

Se llevará a cabo una revisión exhaustiva y sistemática del estado del arte en análisis de señales EEG, identificando las metodologías existentes, técnicas avanzadas y tendencias actuales en el procesamiento de señales cerebrales. Esta revisión permitirá contextualizar el trabajo y validar las decisiones metodológicas.

Entregable: Informe técnico con el análisis del estado del arte y la fundamentación teórica del proyecto.

2.3. Actividad 3: Desarrollo e implementación de modelos de análisis de EEG (Meses 5-6)

Esta etapa constituye el núcleo del proyecto, donde se desarrollarán e implementarán los modelos de análisis de señales EEG. Se realizará el montaje y configuración de los modelos para diferentes tareas específicas, incluyendo pruebas, validación y optimización de los algoritmos desarrollados.

Entregable: Código fuente de los modelos implementados, documentado y funcional.

2.4. Actividad 4: Documentación de resultados consolidados (Meses 6-7)

En la fase final se consolidarán todos los resultados obtenidos durante el proyecto. Se documentarán los hallazgos, análisis de desempeño de los modelos, conclusiones y recomendaciones. Esta documentación servirá como soporte integral del proyecto.

Entregable: Informe técnico final con la documentación completa de resultados, análisis y conclusiones del proyecto.

3. Resultados

3.1. Actividad 1

Durante esta etapa se desarrolló la conceptualización y el diseño de la metodología de trabajo para el procesamiento de señales EEG, orientada a la clasificación del trastorno por déficit de atención e hiperactividad (TDAH) en población infantil. Este proceso incluyó una revisión técnica focalizada en el análisis de señales EEG, arquitecturas de aprendizaje profundo y esquemas de validación cruzada aplicados a datos neurofisiológicos, lo que permitió establecer un marco metodológico consistente y alineado con los objetivos del proyecto.

Como resultado de este análisis, se definieron y evaluaron diferentes enfoques de modelado. Inicialmente, se consideró una arquitectura Transformer linealizada basada en embeddings de Takens (LinearTakensFormer), debido a su capacidad para capturar dependencias temporales en señales no estacionarias. No obstante, evaluaciones preliminares evidenciaron limitaciones en su capacidad de representación. En consecuencia, se establecieron dos líneas de modelado principales que utilizan la misma base de datos EEG ADHD/Control Children: i) un modelo híbrido que combina EEGNet para la extracción de características espaciales y espectrales con un Transformer Encoder para el modelado de dependencias temporales de largo alcance, el cual sustenta el artículo de investigación presentado en el Anexo 1 (<https://www.mdpi.com/2227-7080/14/7/392>); y ii) un modelo alternativo basado en el uso de un kernel personalizado tipo RBF, cuya formulación mejora el desempeño frente a kernels estándar, documentado en el Anexo 2. Ambas aproximaciones responden directamente al objetivo de mejorar la capacidad de discriminación entre sujetos con TDAH y controles.

En esta fase también se definió la base de datos de trabajo, compuesta por registros EEG de 19 canales adquiridos a 128 Hz bajo el sistema internacional 10-20, correspondientes a sujetos en condición de reposo clasificados en dos grupos (TDAH y control). Asimismo, se estableció una cadena de preprocesamiento que incluye filtrado pasabanda entre 0.5 y 60 Hz, eliminación de interferencia de red en 48-52 Hz y segmentación mediante ventanas deslizantes con 50% de solapamiento, garantizando consistencia en la preparación de los datos para ambos modelos.

Finalmente, se diseñó el esquema de validación basado en separación por sujetos bajo el enfoque Leave-K-Subjects-Out, implementado mediante validación cruzada estratificada tipo nested group k-fold con $k = 5$, asegurando evaluación generalizable y evitando fuga de información.

3.2. Actividad 2

En esta etapa, actualmente en desarrollo, se ha llevado a cabo una revisión sistemática del estado del arte en el análisis de señales EEG, con énfasis en técnicas de aprendizaje profundo y enfoques híbridos para clasificación en contextos clínicos. Esta revisión ha permitido validar las decisiones metodológicas adoptadas en la Actividad 1, particularmente en relación con el uso de arquitecturas que integran información espacial, espectral y temporal, así como el uso de esquemas de validación robustos basados en separación por sujetos.

Los resultados preliminares evidencian que las arquitecturas híbridas, como la combinación EEGNet y Transformer, presentan ventajas significativas en la modelación de señales EEG complejas, lo cual es consistente con los resultados hasta el momento en el desarrollo experimental asociado al Anexo 1. De manera complementaria, la exploración de kernels personalizados, como el RBF propuesto, se alinea con tendencias recientes orientadas a mejorar la capacidad de generalización en espacios de características no lineales, en concordancia con los avances documentados en el Anexo 2.

3.3. Actividad 3

En el marco del desarrollo e implementación de modelos de análisis de señales EEG, durante esta etapa se consolidó el modelo propuesto denominado *Temporal Attention and Convolutional Tokenization for Interpretable EEG-Based ADHD Identification in Children*, integrando los componentes metodológicos definidos en fases previas y validando experimentalmente su desempeño bajo un esquema de evaluación independiente por sujetos. La arquitectura desarrollada combina un bloque convolucional inspirado en EEGNet para la extracción de representaciones espaciales y espectrales, un operador Transformer Encoder para el modelado de dependencias temporales y un mecanismo de *attention pooling* orientado a mejorar la interpretabilidad del modelo y la agregación de información relevante en la secuencia temporal.

Como resultado del proceso de validación y optimización, el modelo alcanzó un desempeño estable y competitivo en la tarea de clasificación TDAH/control, obteniendo una exactitud a nivel de sujeto de 87.5 %, *recall* de 96.0 % y precisión de 82.8 %, manteniendo además un número reducido de parámetros entrenables. De manera complementaria, se realizaron análisis de interpretabilidad, estudios de ablación y comparaciones estadísticas frente a modelos de referencia reportados en la literatura, evidenciando mejoras en robustez y consistencia del etiquetado bajo diferentes inicializaciones experimentales. Estos resultados dieron lugar a la preparación, validación y posterior envío del artículo científico asociado a una revista especializada. Finalmente, se publicó un artículo de investigación disponible en el siguiente enlace: <https://www.mdpi.com/2227-7080/14/7/392>

3.4. Actividad 4

Esta fase, actualmente en curso, contempla la consolidación y documentación de los resultados obtenidos durante el proyecto. Como parte de este proceso, el código fuente del modelo EEGTAC fue liberado públicamente en <https://github.com/JulianDPastrana/EEGTAC>, complementando la publicación del artículo científico en revista indexada. Esto facilita la reproducibilidad del trabajo y su continuación por parte de otros investigadores.

Como complemento al enfoque de kernel personalizado RBF documentado en el Anexo 2, se exploró un marco de *Multiple Instance Learning* (MIL) aplicado a la misma tarea de clasificación, disponible en <https://github.com/JulianDPastrana/MIL-Project>. Este enfoque modela cada sujeto como una bolsa (*bag*) de épocas de dos segundos, empleando *gated-attention pooling* para agregar las representaciones en una predicción a nivel de sujeto sin requerir etiquetado por segmento. Los experimentos preliminares, evaluados bajo el mismo esquema de validación cruzada estratificada por sujetos (5-fold), reportan una exactitud de 72.7 % con el baseline de votación mayoritaria y 70.2 % con el módulo MIL de atención sobre un backbone pre-entrenado congelado; el entrenamiento extremo a extremo (*end-to-end*) resulta en menor desempeño (62.1 %), lo que sugiere que el módulo de atención se beneficia de representaciones pre-aprendidas estables.

Estos resultados son preliminares y el desarrollo de esta línea continuará durante los meses restantes del período de vinculación, con énfasis en la comparación sistemática frente a los modelos EEGTAC y RBF y en la exploración de variantes del mecanismo de agregación.

Article Anexo 1

Temporal Attention and Convolutional Tokenization for Interpretable EEG-Based ADHD Identification in Children

Julián David Pastrana-Cortés ^{1,*}, Alejandra Gomez-Rivera ², Andrés Marino Álvarez-Meza ²,
Julian Gil-Gonzalez ¹ and David Cárdenas-Peña ^{1,*}

¹ Automatics Research Group, Universidad Tecnológica de Pereira (UTP), Pereira 660003, Colombia; jugil@utp.edu.co

² Signal Processing and Recognition Group, Universidad Nacional de Colombia, Manizales 170003, Colombia; yeagomezri@unal.edu.co (A.G.-R.); amalvarezme@unal.edu.co (A.M.Á.-M.)

* Correspondence: j.pastrana@utp.edu.co (J.D.P.-C.); dcardenas@utp.edu.co (D.C.-P.)

Abstract

Attention Deficit Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental condition commonly assessed through clinical interviews, behavioral observation, and rating scales. Although electroencephalography (EEG) has emerged as a promising complementary tool for ADHD assessment, robust, subject-independent classification remains challenging due to inter-subject variability, limited datasets, and the need for interpretable computational models. This work introduces EEG-TACT, a compact end-to-end deep learning architecture for identifying ADHD subjects from EEG epochs. The proposed model integrates an EEGNet-inspired convolutional embedding, a Transformer encoder operator, and an attention-based pooling mechanism. Together, these components capture local spatiotemporal EEG patterns, contextual temporal dependencies, and task-relevant latent representations. EEG-TACT was evaluated on a publicly available EEG dataset using strict, subject-independent stratified group partitions, ensuring no data leakage across subjects in the training, validation, and test subsets. Learned temporal filter responses, class-conditioned self-attention maps, and latent-space projections provide model interpretability. An ablation study quantifies the contribution of each architectural component. Performance analysis includes evaluation at the fold, subject, and epoch levels, together with statistical significance comparisons against representative state-of-the-art architectures. EEG-TACT achieved competitive performance among the contrasted models, reaching subject-level accuracy of 87.5%, recall of 96.0%, and precision of 82.8%, while requiring only a few thousand trainable parameters. By exhaustively repeating the initialization, the proposed model demonstrated improved labeling reliability and achieved the best average ranking among the evaluated architectures. The reported results therefore support evidence that EEG-TACT provides a compact, stable, and interpretable model for EEG-based ADHD identification under subject-independent evaluation settings. They also motivate further validation on larger, multi-site, and medication-controlled datasets.



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Keywords: ADHD; EEG; neurodevelopmental disorders; deep learning; convolutional model; Transformer encoder; attention pooling

1. Introduction

Attention Deficit Hyperactivity Disorder (ADHD), one of the most prevalent neurodevelopmental conditions, commonly emerges during childhood with signs such as

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difficulties in academic achievement, social functioning, and emotional regulation [1]. Epidemiological evidence estimates that ADHD affects approximately 5.9% of youth worldwide, with recent meta-analytic estimates reporting prevalence rates of 7.6% in children aged 3–12 years and 5.6% in adolescents aged 12–18 years [2]. Clinically, ADHD is characterized by three main symptom domains: inattention, hyperactivity, and impulsivity, although many children exhibit a combined presentation involving features from more than one domain [3]. Beyond its cognitive and behavioral manifestations, ADHD is frequently accompanied by comorbid conditions such as anxiety, depression, and learning disorders. Therefore, early diagnosis and timely intervention are essential to reduce the long-term negative effects of the disorder on educational performance, social relationships, and overall quality of life [4].

Despite the availability of standardized diagnostic criteria, ADHD assessment remains largely based on clinical interviews, behavioral observation, and parent-, teacher-, caregiver-, or clinician-completed behavioral rating scales [5]. Commonly used instruments, such as the Conners Rating Scales, the Vanderbilt ADHD Diagnostic Rating Scales, and the Swanson, Nolan, and Pelham Rating Scale (SNAP-IV), attempt to quantify inattentive and hyperactive-impulsive behaviors across different environments [6]. Despite their value for structured symptom screening and for collecting multi-informant evidence, the instruments are biased by informant perception, contextual variability, and differences between home and school observations [7]. In addition, behavioral manifestations of ADHD may overlap with other neurodevelopmental, emotional, or learning-related conditions, making differential diagnosis challenging [8]. Consequently, there is a growing interest in complementary objective approaches capable of characterizing neurophysiological patterns associated with ADHD and supporting clinical decision-making without replacing comprehensive clinical evaluation [9].

From a neurobiological perspective, ADHD has been associated with dysfunction in the prefrontal cortex and subcortical structures, particularly the basal ganglia, which are essential for attention regulation, behavioral control, and executive functioning [10].

Given such association with atypical brain function and neurodevelopmental alterations, brain-based neurophysiological and neuroimaging techniques have gained increasing attention as potential objective complementary tools for ADHD assessment [11]. Although functional magnetic resonance imaging has been widely used to characterize functional alterations in ADHD, its use in pediatric populations may be limited by high cost, reduced accessibility, long acquisition times, and sensitivity to motion artifacts [12]. By contrast, electroencephalography (EEG) has emerged as a promising complementary tool for ADHD assessment because it provides a portable, affordable, non-invasive, and high-temporal-resolution means of measuring brain electrical activity [13]. These properties make EEG suitable for characterizing neurophysiological spectral, temporal, and connectivity patterns related to attention, executive functioning, inhibitory control, and behavioral regulation [14].

More recently, the advent of machine learning and deep learning has further advanced the use of EEG for ADHD detection by enabling the analysis of complex, nonlinear, and high-dimensional signal patterns that may not be fully captured through conventional approaches [13]. In line with this, recent studies have shown that EEG-based machine learning approaches can achieve promising performance in distinguishing children with ADHD from typically developing controls, supporting the use of EEG as an objective and data-driven aid for early detection [15].

In this direction, EEG-based studies have explored machine learning strategies to support ADHD diagnosis from subject-level neurophysiological dynamics [16]. Among machine learning approaches, deep learning methods have shown particular potential for

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automatically learning discriminative representations from EEG signals, reducing the dependence on handcrafted feature extraction and enabling end-to-end optimization for EEG classification tasks [17,18]. Convolutional neural networks (CNNs), for example, are effective in capturing local temporal and spatial patterns from EEG data and have been widely used for EEG decoding and visualization [19]. In particular, the EEGNet architecture has emerged as a compact and efficient architecture specifically designed for EEG analysis, thanks to its depthwise and separable convolutions, which extract meaningful features while maintaining a relatively low number of trainable parameters [20]. This makes EEGNet especially attractive for ADHD-related EEG studies, where datasets are often limited and robust feature extraction is essential.

Although CNN-based models are effective at capturing local structures, their finite receptive fields may limit their ability to model long-range dependencies and global contextual information across EEG segments [21]. In this regard, attention mechanisms have gained relevance because they allow the model to focus on the most informative parts of the signal and to capture relationships across distant temporal representations [22]. Building on this idea, Transformer-inspired and hybrid convolutional-Transformer architectures have emerged as promising alternatives for EEG decoding, since they combine local feature extraction with self-attention-based contextual modeling [23]. This type of architecture is particularly relevant for EEG analysis, where discriminative information may be distributed across multiple temporal segments and spatial patterns rather than confined to isolated local features.

This work aims to bridge the gap between compact EEG-specific feature extraction and global sequence modeling for ADHD detection, with global sequence modeling defined here as the modeling of contextual interactions among latent temporal EEG representations within each input epoch, by introducing a novel end-to-end deep learning architecture, termed EEG-TACT. Unlike conventional approaches that rely exclusively on handcrafted features or purely convolutional processing, EEG-TACT combines an EEGNet-inspired convolutional embedding operator with a Transformer encoder to jointly learn local spatiotemporal patterns and short-term temporal dependencies directly from raw EEG signals. The convolutional front-end extracts compact and physiologically meaningful representations across channels and time, while the Transformer module leverages self-attention to model contextual interactions among temporal segments. In addition, an attention-based pooling strategy is incorporated to adaptively emphasize the most informative ADHD EEG representations prior to classification, thereby improving both feature aggregation and interpretability. The proposed model was evaluated on a publicly available EEG dataset composed of school-aged children and assessed using five fixed subject-wise folds under a stratified group cross-validation protocol. The results and comparisons against six state-of-the-art deep learning models show that EEG-TACT provides an effective and compact framework for EEG-based ADHD classification, outperforming several baseline approaches by combining local feature extraction, global dependency modeling, and adaptive attention-based aggregation.

The main contributions of this work are: (i) a compact hybrid convolutional-Transformer architecture that jointly learns frequency-selective spatial filters, short-term attention for contextualizing temporal tokens, and a pooling mechanism that weights the most relevant tokens before classification; (ii) a rigorous evaluation framework combining stratified group cross-validation, varying-seed testing, and subject-independent training to mitigate optimistic performance estimates; and (iii) a multi-faceted analysis encompassing interpretability (parameter visualization, latent-space projections, and ablation experiments) and statistical analysis (paired statistical significance tests, and risk-coverage curves).

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The remainder of this paper is organized as follows: Section 2 describes the EEG dataset, preprocessing pipeline, proposed architecture, training strategy, and subject-independent evaluation protocol. Section 3 reports and discusses the experimental results, including hyperparameter tuning, interpretability analysis, ablation study, comparative performance evaluation, and statistical significance tests. Finally, Section 4 presents the concluding remarks and future research directions.

2. Materials and Methods

This work introduces EEG-TACT, an end-to-end deep learning architecture designed to jointly model local spatiotemporal patterns and long-range temporal dependencies directly from raw EEG signals. The model integrates an EEGNet-inspired convolutional front-end for learning compact spatiotemporal representations from EEG signals with a Transformer encoder that models contextual relationships across temporal segments. Furthermore, an attention-based pooling strategy is employed to adaptively aggregate temporal information, emphasizing the most informative EEG segments for classification. The following subsections present the mathematical formulation of the proposed model in detail.

2.1. Dataset and Preprocessing

This study tests the proposed EEG-TACT using publicly available EEG data collected from children with ADHD and control subjects. The data were shared by Nasrabadi et al. via IEEE DataPort [24]. The database contains EEG recordings from 121 children aged 7 to 12 years. Among them, 61 children were diagnosed with ADHD, and 60 were typically developing controls. ADHD diagnoses were made by an experienced child psychiatrist using DSM-IV criteria [25]. At the time of recording, patients had been using methylphenidate (Ritalin) for up to six months. Thus, the ADHD group corresponded to a clinically diagnosed cohort with reported methylphenidate exposure. The control group had no history of psychiatric or neurological disorders, including epilepsy or high-risk behaviors [25,26].

EEG signals were recorded while participants performed a visually guided attention task designed to assess attentional deficits linked to ADHD [27]. During the task, children viewed cartoon images containing 5 to 16 characters, as depicted in Figure 1a, and were asked to silently count the number of figures in each scene. Images appeared continuously, each new image shown immediately following the participant's response, resulting in variable recording lengths based on individual response speeds [27]. EEG was recorded at a sampling rate of 128 Hz from 19 scalp electrodes placed according to the international 10–20 system (see Figure 1b).

Regarding preprocessing, the EEG signals were first re-referenced to the common average reference, aiming to reduce dependence on the physical recording reference and enhance the consistency of spatial EEG patterns across subjects [28]. This transformation subtracts the instantaneous mean scalp potential from each channel, thereby emphasizing channel-specific deviations from global activity and reducing reference-induced bias. Then, a 50 Hz notch filter attenuates narrow-band power-line contamination, and a 0.5–60 Hz band-pass filter retains the physiologically relevant EEG spectrum while reducing slow drifts, movement-related low-frequency artifacts, and high-frequency muscular or instrumental noise [29]. It is worth noting that after visually inspecting recordings, the subject named *v56p* was removed from further analysis due to marked artifact contamination in the FP1 and FP2 channels, which are strongly associated with alterations in executive, attention, and inhibitory functions [30]. Recordings from remaining subjects were then segmented into 2-second, 50%-overlapping epochs, yielding a (C, T) shaped epochs, where $C = 19$ denotes the number of channels and $T = 256$ the number of temporal samples per

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epoch. This segmentation setting was selected following previous work on the same public ADHD/control EEG dataset, where 2-s windows with 50% overlap were used to increase the amount of temporal information extracted from the recordings while accounting for the variable task duration across subjects [31]. Overall, the preprocessing stage was limited to re-referencing, filtering, and segmenting the continuous EEG recordings into fixed-duration epochs. Consequently, the final dataset preserved the natural inter-subject variability and the epoch-count imbalance present in the original recordings. Finally, the epochs from all retained subjects were concatenated into a global dataset of shape (N, C, T) , where $N = 16,640$ represents the total number of epochs across all subjects, where each epoch inherits the label of its corresponding subject, while subject identifiers were preserved to enable training and evaluation strategy.

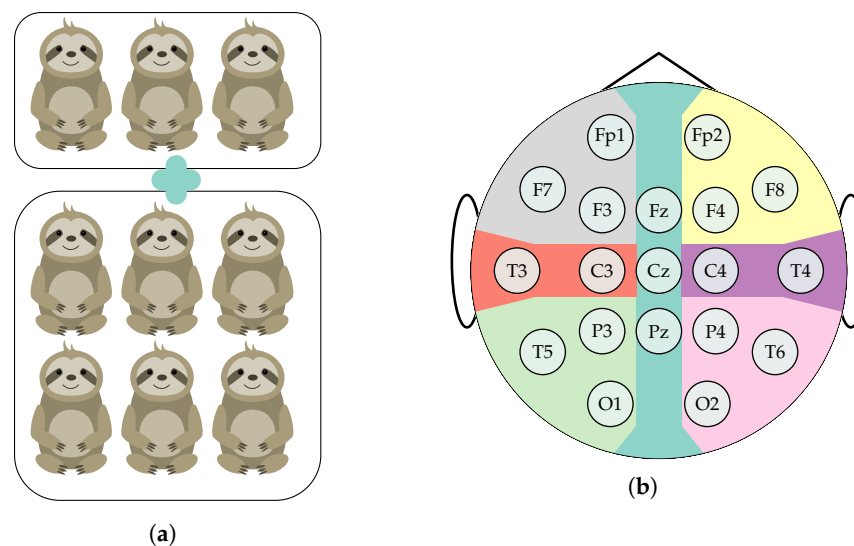


Figure 1. Acquisition setup for EEG-based ADHD assessment. (a) A representative example of the visual counting stimuli presented to participants during recording. (b) The 19-electrode scalp montage arranged according to the international 10–20 system, colour-coded by anatomical region: left frontal, right frontal, midline (Fz, Cz, Pz), left central-temporal, right central-temporal, left posterior, and right posterior.

2.2. EEGNet-Based Feature Embedding

Let $\mathbf{X} \in \mathbb{R}^{T \times C}$ denote a multivariate time-series EEG epoch with C channels and T temporal samples, associated with a label $y \in \mathcal{Y}$ (ADHD: $y = 1$, Control: $y = 0$). The proposed model aims at classifying $\mathbf{X}$ through a parametric mapping $g_{\theta}(\mathbf{X})$, which estimates the posterior probability of belonging to the target class $P(y | \mathbf{X}) \in [0, 1]$. To this end, hierarchical embedding operator extracts informative representations from the input signal, capturing both temporal dynamics and inter-channel dependencies. First, temporal patterns are modeled by applying a set of learnable filters along the time dimension, producing feature maps that encode frequency-selective or time-localized characteristics. Subsequently, channel-wise interactions are incorporated through operations that transform and aggregate information across the spatial (channel) dimension, enabling the model to capture cross-channel dependencies inherent to multivariate signals.

The intermediate representations are further refined using convolutional transformations that jointly model temporal and feature-wise interactions, typically through separable or factorized operations that improve parameter efficiency while preserving expressiveness. These stages include temporal pooling operations, resulting in a downsampled and

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compressed representation with reduced temporal resolution. Formally, the embedding process can be expressed as follows:

$$\mathbf{H}_\varphi = \varphi(\mathbf{X}; \theta_\varphi) \in \mathbb{R}^{T' \times d}, \quad (1)$$

where $\varphi : \mathbb{R}^{T \times C} \rightarrow \mathbb{R}^{T' \times d}$ denotes the embedding operator parameterized by θ_φ , d the number of learned feature components, and T' the temporal instances after transformation and downsampling. In our implementation, each EEG epoch with $T = 256$ samples is transformed into $T' = 8$ latent temporal tokens with embedding dimension $d = 48$, resulting in $\mathbf{H}_\varphi \in \mathbb{R}^{8 \times 48}$. This temporal compression reduces the number of latent tokens processed by the Transformer and the downstream classifier, thereby improving computational and parameter efficiency. Long-range temporal dependencies are then modeled by the Transformer encoder, whose self-attention mechanism allows each latent temporal token to attend to all other tokens in the compressed sequence. The proposed representation encodes the input signal into a compact feature space that captures joint spatiotemporal structure, facilitating downstream classification.

2.3. Transformer-Based Contextual Encoding

Given the embedded representation $\mathbf{H}_\varphi \in \mathbb{R}^{T' \times d}$ resulting from the feature extraction stage, this framework aims to capture long-range temporal dependencies and global contextual interactions across the sequence. To this end, the embedded segment feeds a Transformer-based operator $\mathcal{T}(\mathbf{H}_\varphi; \theta_\tau)$, enabling each temporal position to incorporate information from all other positions in the sequence, resulting in a refined version through self-attention and channel-wise transformations.

The Multi-Head Self-Attention (MHSA) mechanism constitutes the core of the Transformer operator, which models pairwise interactions between temporal tokens [22]. Specifically, the input $\mathbf{H}_\varphi$ is linearly projected into sets of query $\mathbf{Q}_m \in \mathbb{R}^{T' \times d}$, key $\mathbf{K}_m \in \mathbb{R}^{T' \times d}$, and value $\mathbf{V}_m \in \mathbb{R}^{T' \times d}$ matrices:

$$\mathbf{Q}_m = \mathbf{H}_\varphi \mathbf{W}_Q^{(m)}, \quad \mathbf{K}_m = \mathbf{H}_\varphi \mathbf{W}_K^{(m)}, \quad \mathbf{V}_m = \mathbf{H}_\varphi \mathbf{W}_V^{(m)}, \quad (2)$$

with $\mathbf{W}_Q^{(m)}, \mathbf{W}_K^{(m)}, \mathbf{W}_V^{(m)} \in \mathbb{R}^{d \times d}$ as the learnable projection matrices for the m -th self-attention head ($m \in \{1, \dots, M\}$). The attention output for the m -th head results from the weighted average of the values according to the

$$\mathbf{H}_m = \sigma \left(\frac{\mathbf{Q}_m \mathbf{K}_m^\top}{\sqrt{d}} \right) \mathbf{V}_m \in \mathbb{R}^{T' \times d}, \quad (3)$$

being $\sigma(\cdot)$ the softmax activation function that normalizes the relevance of each temporal position with respect to all others. This formulation enables the proposed model to dynamically weight contributions from different time steps, effectively capturing non-local dependencies in the sequence. Then, the concatenation of the head-wise outputs is linearly projected into the multihead self-attention output:

$$\tilde{\mathbf{H}} = [\mathbf{H}_1 | \mathbf{H}_2 | \dots | \mathbf{H}_M] \mathbf{W}_O \quad (4)$$

being $\tilde{\mathbf{H}} \in \mathbb{R}^{T' \times d}$ a contextualized representation where each temporal embedding is expressed as a data-adaptive weighted combination of all embeddings in the sequence, with weights $\mathbf{W}_O \in \mathbb{R}^{(M \cdot d) \times d}$ determined by learned pairwise interactions. Lastly, a residual connection from the EEG embedding allows to preserve the original representation and improves the optimization of the first block $\tilde{\mathbf{H}} = \mathbf{H}_\varphi + \tilde{\mathbf{H}}$.

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Following the attention mechanism, a token-wise N -layered dense block, along with its respective residual connection, enhances the feature expressiveness of the transformer encoder:

$$\mathbf{H}_\tau = \mathcal{T}(\mathbf{H}_\varphi; \theta_\tau) = \tilde{\mathbf{H}} + f(f(\cdots f(\tilde{\mathbf{H}}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \cdots)\mathbf{W}_{N'} + \mathbf{b}_{N'}), \quad (5)$$

where $\mathbf{W}_n \in \mathbb{R}^{F_{l-1} \times F_l}$, $\mathbf{b}_l \in \mathbb{R}^{F_l}$, and $F_l \in \mathbb{N}$ correspond to the trainable weighting matrix, bias vector, and number of neurons at the l -th dense layer, respectively; while $f(\cdot)$ stands for an element-wise non-linear activation function. Further, $\theta_\tau = \{\mathbf{W}_Q^{(m)}, \mathbf{W}_K^{(m)}, \mathbf{W}_V^{(m)}, \mathbf{W}_O, \mathbf{W}_n, \mathbf{b}_n\}_{m,n=1}^{M,N'}$ aggregates the parameters of the full attention encoder to be trained. As a consequence, the attention-based mapping $\mathbf{H}_\tau \in \mathbb{R}^{T' \times d}$ yields a context-aware representation in which each temporal token encodes information aggregated from the entire sequence, effectively enriching the initial embedding $\mathbf{H}_\varphi$ with global temporal structure while preserving its dimensional organization.

2.4. Attention Pooling and Classification Head

Despite unraveling discriminative information for EEG classification, directly operating on the attention embedded trial $\mathbf{H}_\tau$ becomes computationally demanding due to the high dimensionality. Traditional strategies consist of aggregating the sequence into a compact, informative vector representation before classification. However, conventional pooling operators are fixed and non-trainable, limiting their ability to adaptively emphasize the most informative patterns. For that reason, we incorporate an attention pooling mechanism that introduces a learnable aggregation that performs an adaptive temporal aggregation by assigning learnable importance weights to the contextualized tokens and compressing the sequence into a single discriminative vector for classification as follows [32]:

$$\mathbf{z} = \sigma\left(\frac{\mathbf{w}\mathbf{H}_\tau^\top}{\sqrt{d}}\right)\mathbf{H}_\tau, \quad (6)$$

with $\mathbf{w}^\top \in \mathbb{R}^d$ as a learnable parameter vector and $\mathbf{z}^\top \in \mathbb{R}^d$ the resulting discriminative feature vector. Lastly, a classification layer maps $\mathbf{z}$ to the predicted probability:

$$\tilde{p}(y | \mathbf{X}) = \psi(\mathbf{z}; \theta_\psi), \quad (7)$$

where $\psi: \mathbb{R}^d \rightarrow (0, 1)^{|\mathcal{Y}|}$ denotes a feed-forward network parameterized by θ_ψ , followed by a softmax/sigmoid activation. Therefore, the proposed EEG-TACT defines a hierarchical composition of embedding (Equation (1)), transformer (Equation (5)), and classification (Equation (7)) operators, yielding a parametric mapping $g_\theta = \psi \circ \mathcal{T} \circ \varphi$ from an input EEG trial $\mathbf{X}$ to the conditional probability $P(y | \mathbf{X})$, which is learned by optimizing the parameter set $\theta = \theta_\varphi \cup \theta_\tau \cup \theta_\psi$.

2.5. Training Strategy and Evaluation Protocol

To ensure a fair and clinically meaningful evaluation, the model was trained and assessed under a strict, subject-independent evaluation protocol using a stratified group cross-validation scheme. A five-fold stratified group cross-validation scheme was employed, in which each subject was treated as an independent group. This guarantees that data from the same subject never appear in more than one subset among training, validation, and test sets, thereby preventing any form of information leakage, in accordance with best practices in cross-validation and model selection [33]. The stratification procedure preserves the class distribution (ADHD vs. Control) across all folds, while the grouping constraint enforces subject-level separation. At each outer split, subjects were separated

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into training/validation and test subsets. Subsequently, an inner split, also performed using a stratified group strategy, was applied to divide the training portion into training and validation subsets. This hierarchical splitting ensures that model selection and performance estimation remain unbiased with respect to unseen subjects.

During training, the model operates at the epoch level, where each EEG recording is segmented into multiple overlapping windows. Since the ground-truth label is defined at the subject level, a post-processing aggregation step is necessary: for each subject, the predicted probabilities for all their epochs are combined to yield a subject-level prediction. This aggregation enables subject-wise classification from epoch-level outputs. For this work, a majority voting strategy is employed to obtain the final subject-level prediction

$$\hat{y}_s = \underset{i \in \mathcal{I}_s}{\text{mode}} \left[\underset{y \in \mathcal{Y}}{\arg \max} \tilde{p}(y | \mathbf{X}_i) \right], \quad (8)$$

being $\mathbf{X}_i$ the i -th EEG epoch and $\mathcal{I}_s$ the index set of epochs belonging to subject s . The final subject-level prediction $\hat{y}_s$ is therefore defined as the most frequent predicted class among the subject's epochs. This aggregation step ensures that performance metrics are computed at the subject level rather than at the epoch level.

During validation, subject-level accuracy is used as the primary criterion for model selection. This choice is motivated by the nearly balanced class distribution of the dataset and by the fact that accuracy is the main metric reported in the comparative evaluation. Thus, the optimization objective was aligned with the final reporting protocol. During testing, performance metrics are computed by aggregating epoch-level predictions at the subject level. In particular, accuracy, recall, and precision are derived from a single final prediction per subject. This design on training, validation, and testing reflects a realistic clinical scenario in which each subject is assigned a single diagnostic label rather than multiple epoch-level decisions and avoids overestimating performance due to multiple correlated epochs from the same individual.

Finally, hyperparameter optimization was conducted in the inner validation loop using Optuna's Tree-structured Parzen Estimator (TPE) sampler, a sequential model-based Bayesian optimization strategy [34,35]. For each outer fold, a maximum of 20 Optuna trials was evaluated. According to Section 2, the search included hyperparameters of the EEGNet-inspired embedding block, such as the number of temporal filters, depth multiplier, number of separable convolution filters, temporal kernel length, average-pooling factors, activation function, dropout rates, and learning rate. It also included Transformer encoder hyperparameters, comprising the number of attention heads M , the number of Transformer layers, the embedding dimensionality d , the feed-forward dimensionality, and the dense block configuration in Equation (5). Each Optuna trial corresponded to one candidate EEG-TACT configuration, which was trained on the training subset and evaluated on the subject-independent validation subset of the corresponding fold. The optimization objective was the subject-level validation accuracy, computed after aggregating epoch-level predictions by majority voting, in order to align model selection with the final evaluation protocol. Early stopping was applied during model training, and Optuna pruning was used to terminate unpromising trials early, thereby reducing overfitting and computational cost. The best hyperparameter configurations obtained from this search are reported in Table 1 under the corresponding Setting values.

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Table 1. Architecture of the proposed EEG-TACT model. The network comprises a convolutional feature extractor φ , a Transformer encoder $\mathcal{T}$, and a classification head ψ . Tensor dimensions and tuned hyperparameters after the Optuna search routine are reported for each layer. In the Setting column, ELU refers to the Exponential Linear Unit activation function.

Layer	Variable	Output Shape	Hyperparameter	Setting
Feature Extractor φ				
Input epoch	$\mathbf{X}_n$	$C \times T$	C, T	19, 256
Reshape	$\mathbf{X}_{0,n}$	$C \times T \times 1$	–	–
Temporal Conv	$\mathbf{H}_1$	$C \times T \times F_1$	F_1 , kernel	8, (1, 64)
Batch Norm	$\mathbf{H}'_1$	$C \times T \times F_1$	–	–
Depthwise Conv	$\mathbf{H}_2$	$1 \times T \times (F_1 D)$	D , kernel	3, (C , 1)
Batch Norm	$\mathbf{H}^a_2$	$1 \times T \times (F_1 D)$	Activation	ELU
Average Pooling	$\mathbf{H}^p_2$	$1 \times \frac{T}{p_1} \times (F_1 D)$	p_1	4
Spatial Dropout	$\mathbf{H}'_2$	$1 \times \frac{T}{p_1} \times (F_1 D)$	Dropout	0.1
Separable Conv	$\mathbf{H}_3$	$1 \times \frac{T}{p_1} \times F_2$	F_2 , kernel	48, (1, 16)
Batch Norm	$\mathbf{H}^a_3$	$1 \times \frac{T}{p_1} \times F_2$	Activation	ELU
Average Pooling	$\mathbf{H}^p_3$	$1 \times \frac{T}{p_1 p_2} \times F_2$	p_2	8
Spatial Dropout	$\mathbf{H}'_3$	$1 \times \frac{T}{p_1 p_2} \times F_2$	Dropout	0.1
Transformer Encoder $\mathcal{T}$				
Reshape	$\mathbf{H}_\varphi$	$T' \times d$	T', d	8, 48
Transformer Encoder	$\mathbf{H}_\tau$	$T' \times d$	L, M, N' , Dropout	1, 2, 2, 0.3
Classification Head ψ				
Attention Pooling	z	d	–	–
Dropout	$\tilde{z}$	d	Dropout	0.3
Dense classifier	$\tilde{p}$	1	Activation	Sigmoid

3. Results and Discussion

3.1. Model Setup and Hyperparameter Tuning

The proposed EEG-TACT follows a sequential deep learning architecture implemented as an end-to-end model composed of the three main operators defined in Section 2, allowing convolutional feature-extraction capacity, contextual temporal modeling, regularization strength preserving strict subject-independent validation and testing partitions.

Table 1 summarizes the resulting optimal tuned architecture for the proposed EEG-TACT model after Optuna search. The first block resembles an EEGNet-based feature embedding block, applying a temporal convolution to the input time series to extract frequency-selective patterns, followed by batch normalization. Next, a depthwise spatial convolution integrates information across channels and learns spatial filters. The extracted features are then refined through nonlinear activation, average pooling, and spatial dropout. A separable convolution further captures compact temporal representations while reducing the number of trainable parameters. After a second stage of normalization, activation, pooling, and dropout, the resulting feature maps are reshaped into a sequence of latent tokens. The second block corresponds to the Transformer encoder operator processing the embedded representation over temporal tokens. Two self-attention heads and a two-layer dense block exploit long-range temporal dependencies. In the third block, the attention pooling layer aggregates the sequence of encoded tokens using adaptive importance weights. The tuned one-layer Transformer encoder supports epoch classification.

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3.2. Model Interpretability and Ablation Analysis

To provide a comprehensive interpretation of the proposed EEG-TACT, this section analyzes the model from three complementary perspectives. First, the learned parameters are examined at two key layers: the first temporal convolutional filters, which describe how the front-end filters reweight the available EEG frequency content, and the transformer self-attention, which explains the model's distribution of relevance across latent temporal tokens. Second, the evolution of the learned data representation is qualitatively inspected through two-dimensional visualizations that compare the feature spaces at the outputs of the Feature Embedding $\phi(\cdot)$ and the Transformer-Based Contextual Encoding $\mathcal{T}(\cdot)$ operators. Third, an ablation analysis quantifies the contribution of each architectural component by progressively evaluating the effect of temporal self-attention and attention-based pooling under the same subject-independent validation protocol.

Regarding the first temporal convolution, Figure 2 shows the magnitude responses of the learned filters. Taking into account the preprocessing stage, the learned convolutions are interpreted as supervised reweighting of the available spectral content within 4–40 Hz rather than an unconstrained discovery of the full EEG spectrum. As a first insight, the optimal layer uses temporal kernels shared across all 19 EEG channels, making the filters global temporal feature extractors applied uniformly across the 10–20 montage instead of targeting channel-specific or topographical effects, following the EEGNet-inspired principle of first learning temporal filters and then spatial channel representations [20]. At the filter level, narrow attenuations and abrupt variations in the frequency response emerge, indicating that the convolutional stage builds specialized spectral selectors before the subsequent spatial and attention-based layers. This interpretation is consistent with recent EEG decoding frameworks based on filter-bank, multiscale, and explainable temporal convolutional networks, which show that convolutional front-ends can capture task-relevant spectral, temporal, and spectral–spatial EEG representations [36,37]. The qualitative labels in Figure 2 were assigned by inspecting the frequency ranges in which each normalized magnitude response showed relatively stronger components within the analyzed 4–40 Hz passband. F1–F3 and F5 showed stronger responses over the lower-to-intermediate portion of the passband, including theta–alpha and adjacent low-frequency components; F4 and F6 showed noticeable emphasis within the conventional beta range, approximately 13–30 Hz; and F7–F8 exhibited broader responses distributed across a wider portion of the passband. These descriptors are intended to summarize the learned frequency-response profiles and should not be interpreted as strict band-limited physiological decompositions. This behavior is consistent with EEG convolutional and filter-bank-inspired models, where temporal convolutional front-ends can learn complementary frequency-selective transformations from data [20,36]. Since ADHD-related EEG alterations have been associated with activity in theta, alpha, and beta ranges, the learned filters suggest that EEG-TACT exploits complementary spectral patterns rather than a single predefined spectral marker [38].

To interpret the contextual encoding stage of EEG-TACT, average self-attention matrices ($\sigma\left(\frac{\mathbf{Q}_m\mathbf{K}_m^T}{\sqrt{d}}\right)$ from Equation (3)) are extracted from the Transformer encoder on the held-out test set. For each attention head, attention scores are grouped by subject label (ADHD or Control) and averaged across test epochs to obtain class-specific attention maps. Since the Transformer operates over the latent temporal tokens produced by the convolutional embedding block, each attention matrix describes how strongly each encoded temporal segment attends to the others after the EEG signal has already been transformed into a compact spatiotemporal representation. Therefore, these maps describe class-conditioned patterns of interaction among learned temporal EEG representations, following the self-attention principle of modeling pairwise dependencies across sequence elements [22].

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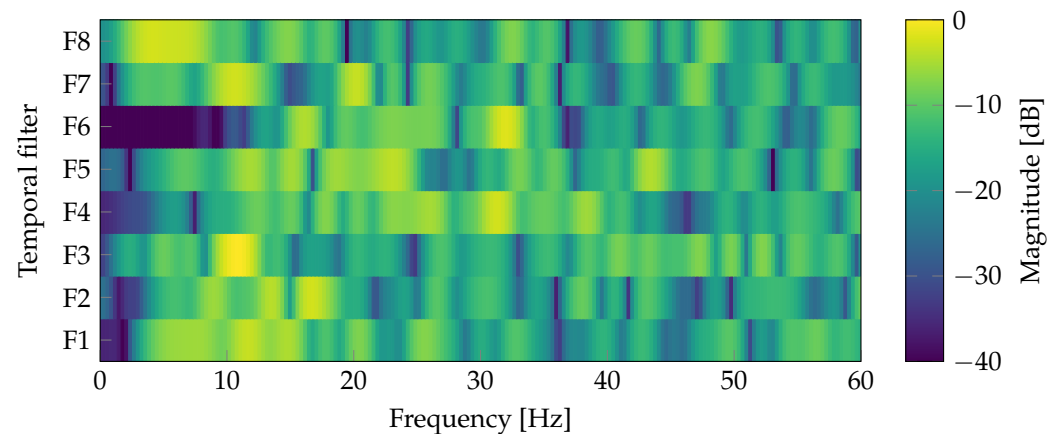


Figure 2. Frequency-domain magnitude responses of the eight temporal convolutional filters learned by EEG-TACT at the first embedding layer. Filters cluster into complementary low-to-mid frequency selectors (F1–F3, F5), beta-range emphasize (F4, F6), and broadband profiles (F7–F8), forming a data-driven filter bank instead, indicating that the convolutional block captures task-relevant spectral diversity.

As shown in Figure 3, the two optimal attention heads (left and right) perform as complementary contextualizations within the transformer. Head 1 (left) shows a more localized structure, with attention concentrated over specific query–key interactions. This behavior is consistent with a discriminative mechanism that emphasizes particular temporal dependencies within the latent sequence. In turn, Head 2 (right) exhibits a more distributed organization, with attention spread across broader regions of the matrix. This pattern suggests a more global temporal encoding mechanism, in which each token integrates information from a wider portion of the embedded sequence. Regarding the discriminative behavior (top to bottom), the average attention maps reveal noticeable differences between classes. The ADHD group (top) exhibits a more structured, higher-contrast organization, with attention weights concentrated in specific query–key regions. This indicates that, during ADHD epochs, the Transformer tends to assign greater relevance to selected interactions among latent temporal tokens rather than distributing attention uniformly across the sequence. Such localized concentration suggests that discriminative information may be carried by specific temporal relationships within the embedded EEG representation, possibly reflecting more irregular or task-relevant temporal dynamics that the model learns to emphasize. In contrast, the Control group (bottom) displays a smoother, more diffuse attention profile, corresponding to a more homogeneous integration of information across temporal tokens. Such a difference suggests that the contextual encoding required to classify Control epochs may rely on broader localized temporal relationships, whereas ADHD epochs induce a more selective attention pattern. Therefore, the self-attention analysis shows that the Transformer operator enables EEG-TACT to distinguish between ADHD and Control by reorganizing local EEG features into class-dependent temporal patterns, combining both localized and distributed attention mechanisms to capture distinctive temporal interactions within each group. Although attention maps should not be interpreted as direct causal or neurophysiological explanations, they provide useful model-level evidence about which latent temporal relationships are emphasized during classification [39,40].

Beyond individual learned parameters, Figure 4 illustrates how the internal token representation evolves across the EEG-TACT pipeline by applying a 2D t-Distributed Stochastic Neighbor Embedding (t-SNE) projection [41]. As an unsupervised nonlinear dimensionality-reduction technique, t-SNE allows inspecting the organization of the learned feature space, rather than the class separability. Each panel uses scatter trans-

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parency to indicate confidence in epoch labeling. The top panel shows substantial overlap between ADHD and Control tokens in the feature embedding $\mathbf{H}_\varphi$. This indicates that the convolutional embedding extracts signal structure, but is only partially separable before contextual temporal modeling. After the Transformer encoder, the t-SNE projection of $\mathbf{H}_\tau$ (bottom left) organizes the two classes more clearly. Additionally, the tokens near the opposing class exhibit the lowest labeling confidence, indicating an awareness of uncertainty under potential mistakes. These elements suggest that self-attention helps weight temporal interactions, as seen in Figure 3, and reshapes the latent representation into a more discriminative geometry. The bottom-right panel colors samples by subject, offering another view of latent organization. Epochs from a single subject form local neighborhoods or coherent projected trajectories, because EEG epochs from the same participant share neurophysiological and recording signatures. Such insight supports the use of subject-grouped cross-validation. This approach avoids information leakage between the training and test sets [42]. Hence, EEG-TACT guides convolutional features toward stronger class organization while capturing subject-specific patterns.

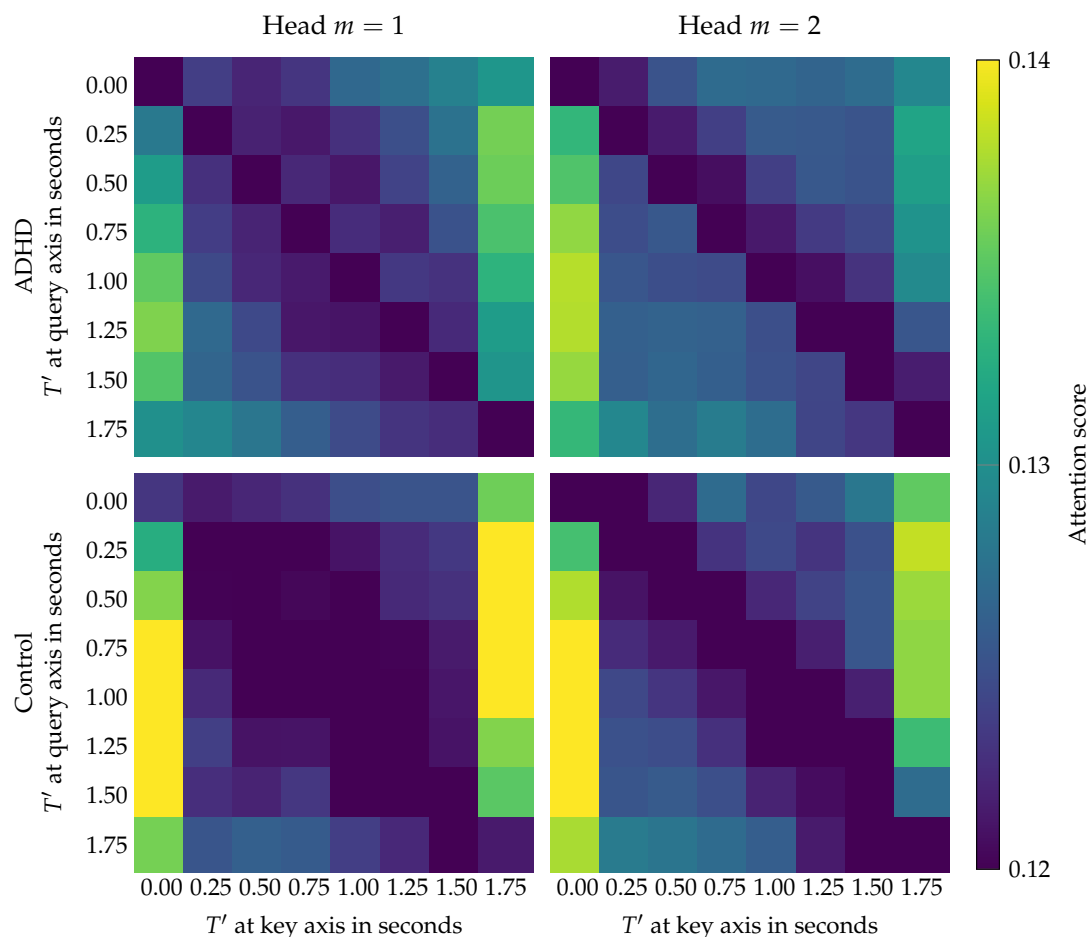


Figure 3. Average of self-attention matrices $\sigma\left(\frac{\mathbf{Q} * m \mathbf{K} * m^T}{\sqrt{d}}\right)$ of the Transformer encoder on the test set, computed from epochs of the held-out test-set subjects across the five subject-independent folds, for each class and attention head. Key and query axes represent latent temporal tokens T' within the epochs, expressed in seconds. Subjects with ADHD produce more contrasted patterns, while controls show a more diffuse distribution among tokens. Heads complement more localized and distributed patterns.

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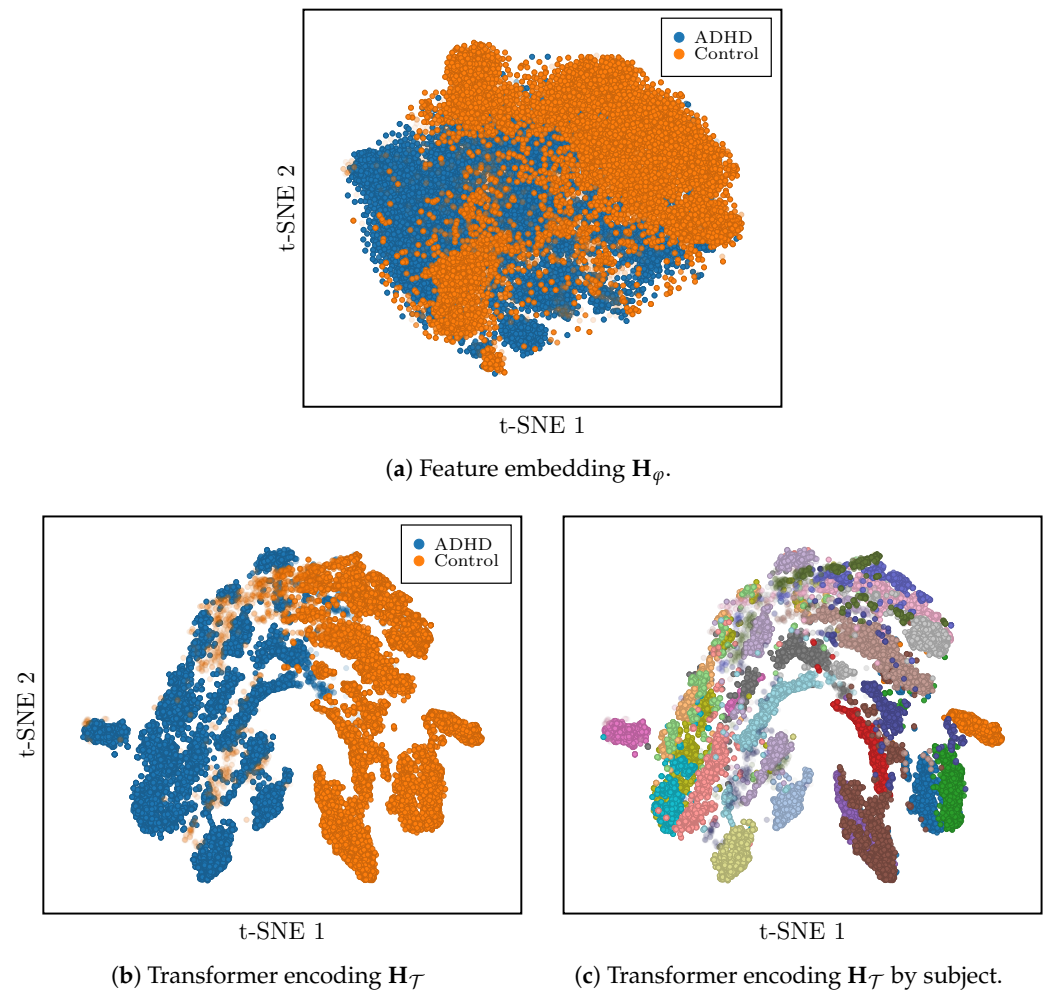


Figure 4. t-SNE visualization of the learned EEG representations for the temporal tokens at (a) the feature embedding H_ϕ (a) and (b,c) after the transformer encoding H_T . (a,b) identify classes by color. (c) color codes tokens according to the subject. For all panels, scatter transparency is proportional to the epoch labeling confidence of EEG-TACT.

The last interpretability perspective concerns the contribution of the main architectural components. To this end, an ablation study progressively evaluates the effects of temporal self-attention and attention-based pooling under a common validation protocol. This design ensures that the observed differences are attributable to architectural changes rather than to data-splitting variability, since all configurations are assessed using the same subject-grouped, five-fold stratified cross-validation protocol, a fixed random seed, and identical subject-wise partitions. The ablation design evaluates four configurations: an EEGNet-like baseline using flatten-based aggregation [20], an EEGNet variant equipped with attention pooling, a Transformer-enhanced variant using global average pooling [22], and the complete EEG-TACT architecture combining the Transformer encoder with attention-based aggregation. In this setting, the Transformer block evaluates the benefit of temporal self-attention for contextual sequence modeling, whereas the pooling strategy evaluates how temporal features are summarized before classification.

Table 2 reports the mean fold-wise performance of each configuration. The results show a consistent improvement from the most straightforward configuration to the complete proposed model, supporting the complementary roles of temporal contextualization and adaptive aggregation. The EEGNet-like baseline provides the lowest overall performance, indicating that flatten-based aggregation has limited capacity to summarize temporally distributed EEG features in a subject-independent setting. Then, introducing

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attention pooling slightly improves classification, suggesting that learnable aggregation can help emphasize informative temporal segments even when the preceding representation is generated only by convolutional operations. Adding the Transformer block while using global average pooling further improves performance and reduces fold-to-fold variability, indicating that self-attention provides a more stable and discriminative temporal representation by modeling dependencies across embedded EEG segments. Nevertheless, the complete EEG-TACT outperforms all previous ablated configurations by providing a task-adaptive attention pooling to weight the most relevant temporal representations before classification. These results support the architectural premise that the synergistic contributions of convolutional feature extraction, temporal self-attention, and attention-based pooling improve the identification of ADHD from EEG signals.

Table 2. Progressive ablation analysis under subject-grouped, five-fold stratified cross-validation protocol using a fixed random seed. Accuracy, recall, and precision are reported as mean $\pm$ standard deviation (%) across folds. Boldface indicates the best-performing configuration.

Configuration	Architecture		Performance (%)		
	Transformer	Pooling	Accuracy	Recall	Precision
EEGNet-like Baseline	$\times$	Flatten	80.8 $\pm$ 8.1	91.0 $\pm$ 12.6	76.8 $\pm$ 9.6
Baseline + Attention Pooling	$\times$	Attention	81.7 $\pm$ 7.6	91.5 $\pm$ 14.4	78.3 $\pm$ 10.7
Baseline + Transformer	$\checkmark$	Global Average	82.5 $\pm$ 4.6	90.8 $\pm$ 9.6	78.8 $\pm$ 7.3
EEG-TACT (Proposal)	$\checkmark$	Attention	87.5 $\pm$ 5.1	96.0 $\pm$ 8.9	82.8 $\pm$ 4.0

Complementarily, Figure 5 presents the subject-level epoch classification obtained by the four configurations evaluated in the ablation study. In each panel, vertical bars correspond to subjects and summarize the percentage of epochs assigned to the ADHD (blue) and Control (orange) classes. The dashed horizontal line indicates the 50% majority-vote threshold used to convert epoch-level predictions into a final subject-level decision (please refer to the Supplementary Material for a more detailed epoch-level evaluation). Subjects are ordered to emphasize classification confidence: true ADHD subjects are placed on the left and sorted from the highest to the lowest percentage of ADHD votes, whereas true Control subjects are placed on the right and sorted from the highest to the lowest percentage of Control votes. Therefore, correctly classified subjects appear as bars dominated by the expected class color on each side of the plot, while subjects near the center represent the most ambiguous cases. The pale central band highlights misclassified subjects; consequently, its width provides a direct visual indication of the subject-level classification error for each configuration.

As in Table 2, a positive quality progression is observed when introducing the architectural components. The EEGNet-like baseline exhibits the least favorable epoch-labeling profile, with a wider pale central band and several subjects near the 50% decision threshold, indicating that flatten-based aggregation produces less reliable subject-level evidence. Despite improving the baseline by reducing epoch mislabeling, adding the attention pooling still presents a persistent misclassification band, indicating that adaptive pooling alone is insufficient when the temporal representation is generated solely by convolutional operations. In turn, incorporating the Transformer block with global average pooling further improves the epoch-labeling structure, as the correct-class percentages decrease more gradually toward the center and fewer subjects fall into the mislabeled region, supporting the role of self-attention in producing more coherent contextual representations across epochs. Lastly, the complete EEG-TACT yields the most favorable pattern: the pale band is narrower, many correctly classified subjects retain high true-class vote percentages, and the remaining errors are less dominated by mislabeled epochs. Such a labeling quality progression reinforces the ablation findings by showing that the combination of Transformer-based temporal

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contextualization and attention-based pooling leads to more reliable majority-vote behavior than either component alone, and indicating that EEG-TACT improves both the number and the quality of subject-level decisions.

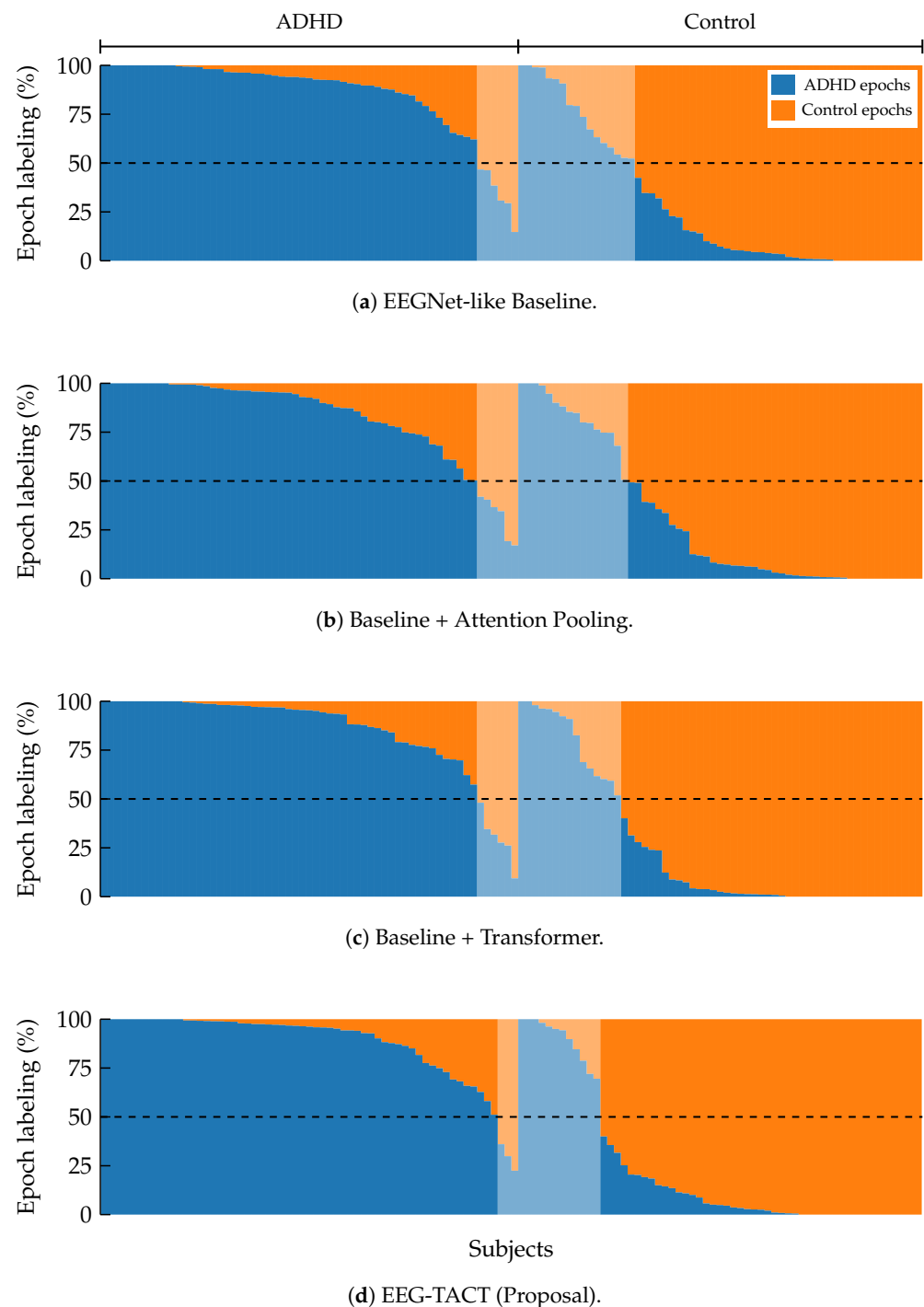


Figure 5. Subject-wise epoch-labeling profiles for the ablation study under the stratified five-group cross-validation protocol. Each panel corresponds to one compared configuration. For each subject, stacked bars show the percentage of epochs classified as ADHD and Control. True ADHD subjects are displayed on the left and sorted from highest to lowest ADHD-vote percentage. True Control subjects are displayed on the right and sorted from highest to lowest Control-vote percentage toward the center. The pale central band indicates misclassified subjects.

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3.3. Performance and Significance Analysis

For comparative performance evaluation, the proposed EEG-TACT is benchmarked against representative EEG deep-learning architectures previously used for EEG decoding and ADHD-related classification tasks. EEGNet is included as a compact EEG-specific convolutional network that combines temporal convolutions, depthwise spatial filtering, and separable convolutions to efficiently learn frequency-specific spatial representations with a reduced number of trainable parameters [20]. The architecture of the ShallowConvNet, inspired by the filter-banked common spatial patterns, emphasizes band-power modulations via temporal and spatial convolutions, followed by squaring, average pooling, and logarithmic activation [19]. CNN-LSTM combines convolutional layers for local spatiotemporal feature extraction with recurrent LSTM units to model longer temporal dependencies in ERP-related ADHD responses [43]. In addition, the residual convolutions of the Multi-Stream model exploit skip connections to stabilize deeper temporal feature learning and mitigate degradation or vanishing gradient effects [31]. The design of the EEG Conformer, which similarly combines Convolutional and Transformer blocks. Although at first look it might be similar to the proposed model, convolutional does not include depth-wise and point-wise features, and the aggregation mechanism relies on average pooling [21]. The integrated IM-CBGT combines convolutional layers for local feature extraction, bidirectional LSTM units for modeling forward-backward temporal dependencies, and GRU-Transformer blocks with multi-head self-attention to capture global sequence relationships before final classification [44]. Finally, T-GARNet constitutes a recent ADHD-oriented architecture that integrates Transformer-based temporal attention, multi-scale Gaussian-kernel functional connectivity, and Rényi-information regularization to capture long-range temporal dependencies, spatial interactions among channels, and interpretable connectivity patterns [45].

All state-of-the-art models are trained and validated under the same experimental conditions as EEG-TACT described in Section 2.5, including identical preprocessing, subject-wise data partitioning, Optuna's setup search, training criteria, and subject-level performance estimation, thereby ensuring that the reported differences reflect architectural and representational factors rather than protocol-dependent biases. Additionally, each model is validated using a varying-seed, subject-independent protocol. First, five fixed subject-wise folds are generated using stratified group cross-validation. Then, the training procedure is repeated ten times with different random seeds, yielding 50 test evaluations. This design provides a more robust and reproducible estimate of model performance than a single-run evaluation, since it accounts not only for variability across unseen subject partitions but also for stochastic effects arising from weight initialization, mini-batch ordering, dropout, and optimizer dynamics.

Table 3 summarizes the comparative performance of the models across two complementary statistical settings: the varying-seed evaluation over 50 test outcomes and a subject-level matched comparison using the best seed. The varying-seed setting reports the mean and standard deviation across the five fixed stratified group folds and the ten independent random seeds, while the best-seed setting averages the five folds from the best initialized model. The reported metrics include Accuracy, Recall, Precision, Specificity and Area Under the Curve (AUC). Each setting is supported with a statistical test. For the varying-seed analysis, the Wilcoxon signed-rank test is used to evaluate statistical differences between EEG-TACT and each baseline over the 50 paired scores, without assuming a parametric form for the distributions of the reported metrics [46]. We also report the Average Rank based on Accuracy, computed as the mean ranking position of each model after sorting the models according to their Accuracy scores. For the latter, the McNemar test validates the significance of the odds ratio for paired subject-level decisions, that is,

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whether the number of subjects correctly classified by EEG-TACT but misclassified by the baseline differs significantly from the number of subjects correctly classified by the baseline but misclassified by EEG-TACT [46]. Filled cells indicate statistically significant differences ($p < 5\%$) relative to EEG-TACT, determined using the Wilcoxon or McNemar test.

Table 3. Performance statistics for contrasted models under ten varying seeds (top) and at the best seed (bottom). Metrics are reported as mean $\pm$ standard deviation. Bold-face indicates the best model in the row. For varying seed, filled cell indicates significant difference ($p < 5\%$) between EEG-TACT and the baseline using the Wilcoxon signed-rank tests on paired scores. In the odds ratio row, filled cell indicates significant difference ($p < 5\%$) according to the McNemar test.

Model	EEGNet	ShallowConvNet	CNN-LSTM	Multi-Stream	EEGConformer	IM-CBGT	T-GARNet	EEG-TACT
Year	2017	2017	2022	2023	2023	2024	2025	2026
Parameters	1746	36,522	9052	574,082	789,506	1,195,266	9071	7322
Varying seed (average over 50 folds)								
Accuracy (%)	80.7 $\pm$ 1.3	78.7 $\pm$ 3.2	84.2 $\pm$ 7.4	69.3 $\pm$ 2.9	77.2 $\pm$ 8.2	70.6 $\pm$ 9.6	74.8 $\pm$ 2.9	84.2 $\pm$ 1.8
Recall (%)	88.8 $\pm$ 2.0	84.9 $\pm$ 12.2	93.8 $\pm$ 8.9	79.3 $\pm$ 3.6	89.8 $\pm$ 12.8	80.0 $\pm$ 12.0	96.2 $\pm$ 1.3	92.8 $\pm$ 2.6
Precision (%)	77.7 $\pm$ 1.2	78.9 $\pm$ 5.3	79.9 $\pm$ 7.7	67.9 $\pm$ 3.3	73.6 $\pm$ 8.5	68.9 $\pm$ 9.6	68.3 $\pm$ 2.1	80.0 $\pm$ 1.6
Specificity (%)	74.2 $\pm$ 1.7	74.0 $\pm$ 9.2	75.1 $\pm$ 11.0	60.2 $\pm$ 4.7	65.3 $\pm$ 13.9	61.6 $\pm$ 15.0	53.4 $\pm$ 6.0	76.2 $\pm$ 2.2
AUC (%)	89.2 $\pm$ 1.0	90.3 $\pm$ 0.8	92.0 $\pm$ 5.6	76.3 $\pm$ 2.1	86.5 $\pm$ 7.2	78.9 $\pm$ 5.0	78.6 $\pm$ 3.8	90.5 $\pm$ 1.3
Average Rank	3.85	4.12	2.65	6.60	4.78	6.10	5.29	2.61
Best seed (average over 5 folds)								
Accuracy (%)	79.2 $\pm$ 9.3	80.8 $\pm$ 3.7	80.0 $\pm$ 10.8	73.3 $\pm$ 5.6	74.2 $\pm$ 9.0	69.2 $\pm$ 9.6	70.0 $\pm$ 14.3	87.5 $\pm$ 5.1
Recall (%)	88.4 $\pm$ 16.5	91.1 $\pm$ 9.0	93.3 $\pm$ 14.9	79.1 $\pm$ 13.4	85.2 $\pm$ 23.5	77.4 $\pm$ 11.0	96.8 $\pm$ 4.4	96.0 $\pm$ 8.9
Precision (%)	76.2 $\pm$ 8.5	76.7 $\pm$ 8.2	76.0 $\pm$ 11.3	74.1 $\pm$ 12.8	73.2 $\pm$ 10.5	68.4 $\pm$ 10.4	64.8 $\pm$ 12.2	82.8 $\pm$ 4.0
Specificity (%)	71.9 $\pm$ 9.7	71.9 $\pm$ 8.0	68.8 $\pm$ 17.6	68.2 $\pm$ 19.1	64.7 $\pm$ 20.1	61.0 $\pm$ 18.7	43.2 $\pm$ 25.6	79.6 $\pm$ 3.5
AUC (%)	87.2 $\pm$ 6.3	91.2 $\pm$ 3.4	92.1 $\pm$ 7.2	78.0 $\pm$ 8.7	84.6 $\pm$ 8.4	78.5 $\pm$ 4.8	72.1 $\pm$ 14.2	89.9 $\pm$ 5.2
Odds ratio	11/1	11/3	11/2	24/7	19/3	28/6	23/2	–

In the first setting, EEG-TACT achieves the most robust overall performance, only reached by the T-GARNet Recall and the CNN-LSTM Accuracy and AUC. Nonetheless, the smaller standard deviation's Accuracy of EEG-TACT indicates that the proposal is less sensitive to stochastic training factors such as weight initialization, mini-batch ordering, dropout, and optimizer dynamics. This stability is further reflected in its best average rank across accuracy scores, suggesting that the proposed model performs consistently well across repetitions rather than depending on a favorable seed. In terms of class-wise performance, EEG-TACT achieves the highest precision and Specificity while maintaining high recall and AUC, yielding a more balanced discrimination between ADHD and control. By contrast, T-GARNet, EEGConformer and CNN-LSTM obtain very high recall but noticeably lower precision, suggesting a stronger tendency to classify subjects as ADHD and therefore to increase false positives. The Wilcoxon tests confirm the significance of the improvement against EEGNet, ShallowConvNet, Multi-Stream, EEGConformer, IM-CBGT, and T-GARNet. It should be noted that the significant difference in the Recall score between EEG-TACT and T-GARNet favors the latter.

In the best-seed setting, EEG-TACT achieves the highest subject-level accuracy precision, and specificity, while maintaining one of the highest recall values, indicating that under the most favorable initialization, EEG-TACT not only detects ADHD subjects effectively but also better controls false positive ADHD predictions than the competing models. Such a dissimilarity is particularly relevant because several baselines, especially T-GARNet and CNN-LSTM, reach comparable or even slightly higher recall but at the cost of substantially lower precision and specificity, reflecting a less balanced diagnostic behavior. Further, the odds ratios support the subject-level superiority of EEG-TACT. For example, the 11/2 odds ratio against CNN-LSTM indicates that EEG-TACT correctly classified eleven subjects missed by CNN-LSTM, whereas CNN-LSTM corrected only two subjects missed by EEG-TACT. The McNemar test reveals several statistically significant differences, indicating that the improvement is not only reflected in aggregate metrics but also translates into concrete gains in subject-level decisions. As a general finding, EEG-TACT achieves a favorable accuracy–reliability–complexity trade-off, demonstrating that performance advantages

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arise from an effective EEG-specific architectural bias rather than increased model size, and supporting its suitability for robust subject-independent ADHD detection.

Due to the majority voting scheme for subject labeling, a correct subject-level decision may still arise from a mixture of confident correct and confident incorrect epoch predictions. To assess prediction reliability, risk-coverage analysis addresses this by evaluating selective classification behavior at the epoch level [47]. Selective risk measures the error rate over the most confidence retained epochs, while coverage denotes the proportion of epochs included. Figure 6 illustrates the coverage-risk curves for each model, where a well-calibrated classifier is expected to produce low selective risk at low coverage levels and a gradual increase as coverage approaches 100%. The risk-coverage curves reveal marked differences in the confidence-ranking behavior of the evaluated models. Among the baselines, Multi-Stream and IM-CBGT exhibit substantially higher selective risk across nearly the entire coverage range, which is associated with poor confidence calibration and a larger fraction of high-confidence errors. The non-monotonic growing of T-GARNet, with a sharp increase in risk at low coverage and a partial decrease at intermediate coverage, implies that the model may assign excessive confidence to a small number of incorrect predictions. Remaining models exhibit a typical evolution: their risk remains relatively low at low coverage values but increases more rapidly as coverage increases, suggesting weaker separation between reliable and unreliable epochs. Finally, EEG-TACT yields the lowest selective risk, with the smoothest risk increment over coverage. Therefore, the proposed model provides confidence estimates with a reliable ordering of easy and difficult trials, demonstrating that epoch-level predictions are not only more accurate but also better ordered by confidence, yielding a more favorable selective classification profile under uncertainty.

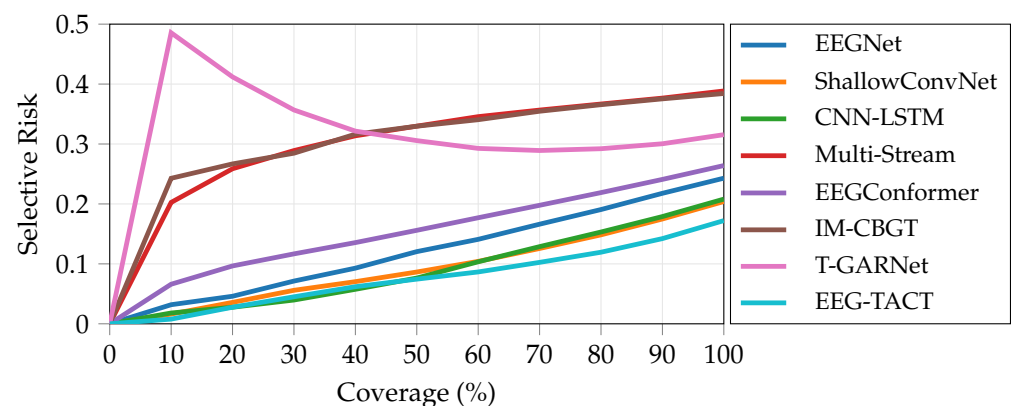


Figure 6. Selective risk versus coverage for the evaluated models using the best-performing seed. Coverage levels are evaluated every 10%. Selective risk is computed as the error rate over the retained epochs. EEG-TACT achieves the lowest selective risk and the smoothest increase across the coverage range, indicating improved confidence calibration and a more reliable ranking of easy and difficult trials under uncertainty.

4. Concluding Remarks and Future Work

This work introduces EEG-TACT, a compact end-to-end architecture for subject-level identification of ADHD from EEG epochs by integrating convolutional feature extraction, transformer-based temporal contextualization, and attention-based aggregation into a single differentiable framework. This design aims to learn frequency-selective temporal filters, spatial channel representations, contextual dependencies among latent temporal tokens, and adaptive temporal summaries for classification. The improvement on EEG decoding under clinically meaningful subject-level evaluation validates the competitive, statistically robust performance of EEG-TACT while maintaining a reduced number of

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trainable parameters, indicating that its performance gains arise from an appropriate architectural bias rather than increased model complexity alone.

The main findings support the relationship between convolutional feature extraction, temporal self-attention, and adaptive pooling, which improved both representation learning and subject-level classification performance. The learned temporal filters work as data-driven spectral reweighting mechanisms within the preprocessed EEG band, capturing complementary low-, mid-, and beta-range patterns. In addition, the self-attention maps highlight class-dependent temporal interactions among latent tokens, while the latent-space projections reveal local EEG feature reorganization from the Transformer-based contextual encoding into a more discriminative representation. The ablation analysis further confirmed that neither temporal self-attention nor attention pooling alone fully explains the observed improvement. Instead, the complete EEG-TACT configuration provided the most favorable subject-level labeling profiles.

From a methodological and practical perspective, this study reinforces the importance of evaluating EEG-based ADHD classifiers under strict subject-independent protocols. By using stratified group partitions, nested validation, and statistical testing, the proposed framework reduces the risk of information leakage and provides a more realistic estimate of model behavior on unseen participants. These aspects are particularly relevant for translational EEG research, where a model should not only achieve high accuracy but also produce stable, interpretable, and confidence-consistent predictions. Therefore, EEG-TACT has the potential to serve as a computational decision-support approach to complement comprehensive clinical assessment.

Despite the promising and reliable results, the following identified limitations offer future research directions. First, the model was trained at the epoch level, requiring post hoc majority voting for subject labeling and constraining the time-horizon analysis. From a methodological perspective, new experiments may include efficient attention mechanisms, temporal conformer-like modules, hierarchical transformers, spatial-temporal attention schemes and epoch aggregation techniques to integrate whole-recording classification and improve long-range dependency modeling while preserving computational efficiency [48]. Second, the model was evaluated on a single public binary ADHD/control EEG dataset, which limits conclusions about generalization across recording conditions (e.g., multi-site datasets, electrode montages, and recording devices) and clinical populations (e.g., ADHD subtypes, comorbidities, medication effects, and longitudinal clinical variability). Also, medication status should be considered when interpreting the present results, since ADHD participants had reported methylphenidate exposure at the time of recording. According to the Ritalin LA prescribing information, methylphenidate has a short pharmacokinetic profile in children, with an average half-life of approximately 2.5 h, and no accumulation is expected following multiple once-daily oral dosing [49]. However, the dataset does not provide dosage, formulation, time since last administration, washout condition, or pre/post-medication EEG recordings. Thus, it is not possible to disentangle ADHD-related EEG signatures from potential medication-related effects, especially considering that medication status may modulate EEG or ERP markers in children with ADHD [50]. Lastly, future work should not only externally validate EEG-TACT under varying protocol settings but also prioritize extending the architecture to support heterogeneous EEG acquisition conditions and multiple tasks toward a clinical foundation model, while incorporating systematic artifact-detection pipelines with quantitative and reproducible criteria for trial- or subject-level exclusion [51].

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/technologies14070392/s1>, Subject-wise epoch-labeling profiles for the ablation study under the stratified five-group cross-validation protocol.

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RBF Conformer for Dummies

L. T. Castañeda-Rojas, J. D. Pastrana-Cortés, and
D. Cárdenas-Peña

Automatics Research Group, Universidad Tecnológica de Pereira

April 29, 2026

1 Abstract

2 Introduction

3 Mathematical Framework

3.1 Problem Setting

Given a set of subjects, each associated with a variable number of EEG trials, let $\mathcal{D} = \{(\mathbf{X}_{s,n}, y_{s,n})\}_{s=1, n=1}^{S, N_s}$ denote the dataset, where S is the total number of subjects and N_s is the number of trials for subject s . Each trial $\mathbf{X}_{s,n} \in \mathbb{R}^{C \times T}$ represents an EEG recording with C channels and T temporal samples, and $y_{s,n} \in \{0, 1\}$ denotes its corresponding label. Notably, all trials belonging to a given subject share the same label.

Given an input trial $\mathbf{X}_{s,n}$, the model outputs a probability $\hat{y}_{s,n} \in (0, 1)$ of belonging to label 1. The objective is to learn a mapping that accurately predicts the labels across all trials while accounting for the subject-wise structure of the data. In the same spirit of [1], the proposed model is structured into three modules designed to learn both local and global features across the channel and temporal dimensions.

3.2 Embedding Module-**propuesta**

Given an input trial $\mathbf{X}$, and omitting subscripts for simplicity, a bank of F filters with kernel size $(1, k_t)$ is applied in a channel-wise manner, producing a set of temporally filtered representations of the EEG trial. Subsequently, to integrate information across channels, a convolution with kernel size $(C, 1)$ is applied across all channels for each filter.

This process yields an embedding representation $\mathbf{Z} \in \mathbb{R}^{F \times T'}$, where F denotes the number of learned filters and defines the dimensionality of the latent

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feature space, while T' corresponds to a reduced temporal resolution resulting from the convolutional operations. In this sense, T' can be interpreted as a downsampled version of the original temporal dimension, preserving the most relevant patterns while reducing temporal redundancy.

This formulation produces a compact representation of the EEG trial, where the original signal is projected onto a set of learned filters that capture discriminative spatio-temporal patterns. The reduced temporal resolution further lowers the computational cost of subsequent transformer layers while retaining the essential signal dynamics.

3.3 Transformer Encoder Module

From the embedding representation $\mathbf{Z}$, the model employs a transformer encoder to capture long-range dependencies between temporal segments of the EEG signal [2]. This module enables the modeling of global interactions between the embedded representations generated by the convolutional stage, overcoming the limitations of purely local methods. Each layer of the transformer encoder consists of two main components: a multi-head self-attention mechanism and a position-wise feed-forward network.

3.3.1 Self-Attention

Given the embedding representation $\mathbf{Z} \in \mathbb{R}^{F \times T'}$, the model projects it into three distinct subspaces, yielding the queries, keys, and values matrices:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{Z}, \quad \mathbf{K} = \mathbf{W}_K \mathbf{Z}, \quad \mathbf{V} = \mathbf{W}_V \mathbf{Z}, \quad (1)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times F}$ are learnable projection matrices, and D corresponds to the dimensionality of each subspace. In the standard formulation, the self-attention mechanism operates on the queries, keys and values matrices via scaled dot-product operation as follow in Equation (2):

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\mathbf{Q}^\top \mathbf{K}}{\sqrt{D}} \right) \mathbf{V}^\top \in \mathbb{R}^{T' \times D}, \quad (2)$$

with $\text{Softmax}(\cdot)$ operator applied row-wise to produce normalized attention weights across the temporal dimension.

3.3.2 Multi-Head Attention

In the multi-head formulation, self-attention process described in Equation (2) is repeated across H independent attention heads, each one producing an output denoted as $\text{head}_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i)$. Then, the outputs of all attention heads are then concatenated along D dimension:

$$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\text{head}_1 \parallel \dots \parallel \text{head}_H] \mathbf{W}_O \in \mathbb{R}^{T' \times F}, \quad (3)$$

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with $\mathbf{W}_O \in \mathbb{R}^{F \times F}$ as output projection matrix. In this way, each head capture different temporal dependencies and integrate them with a linear projection. However, the above formulation impose the constraint $D = \frac{F}{H} \in \mathbb{N}$, i.e, the number of filters F must be a multiple of the number of heads H . To overcoming this restriction, the embedding representation $\mathbf{Z}$ is first linearly projected into a suitable embedding space through a learnable transformation matrix before MHA operation.

3.3.3 Position-Wise Feed-Forward and Classification Head

Following the multi-head self-attention mechanism, the resulting representation undergoes processing by a position-wise feed-forward network applied independently to each temporal position. Additionally, residual connections are further incorporated around both the multi-head attention and feed-forward sublayers. The above constitute a single transformer encoder layer, which can be stacked to an arbitrary numbers of layers.

For the classification task, the output of the last transformer encoder layer is flattened into a one-dimensional vector and passed through a multi-layer perceptron. Finally, a sigmoid activation the produces the probability to EEG trial of belonging to class 1, i.e., $P(\hat{y} = 1 \mid \mathbf{X})$.

3.4 Kernel-Based Transformer Encoder -Proposed

Starting from the formulation given in Equation (2), we adopt a generalized similarity-based framework in which the interaction between queries and keys is defined through a kernel function, rather than being restricted to the standard dot-product [3]. The attention operator is defined in a row-wise manner, where each i -th row corresponds to the output given by Equation (4).

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_i = \sum_{j=1}^{T'} \frac{k(\mathbf{Q}_i, \mathbf{K}_j)}{\sum_{\ell=1}^{T'} k(\mathbf{Q}_i, \mathbf{K}_\ell)} \mathbf{V}_j \in \mathbb{R}^D, \quad (4)$$

where $\mathbf{Q}_i, \mathbf{K}_j, \mathbf{V}_j \in \mathbb{R}^D$ denote the i -th and j -th row vectors of the matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V}$, respectively, as defined in Equation (1). The function k represents a kernel that quantifies pairwise similarity between the query $\mathbf{Q}_i$ and the key $\mathbf{K}_j$. This formulation enables the model to flexibly adapt interactions across temporal elements, thereby enhancing its capacity to capture complex and non-linear relationships.

Since the query, key, and value matrices are obtained as linear projections of the embedding $\mathbf{Z}$ according to Equation (1), each column $\mathbf{z}_j \in \mathbb{R}^F$ represents the feature vector associated with the j -th temporal position. In this sense, the attention mechanism can be reinterpreted directly in the embedding space, where interactions between representations are fully determined by similarity relationships between $\mathbf{z}_i$ and $\mathbf{z}_j$.

Under this perspective, the normalized form of the kernel in Equation (4) naturally induces a normalized weighting scheme over the samples, which can be

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99 interpreted as a conditional distribution over the available representations. In
 100 particular, these weights can be understood as an approximation of the condi-
 101 tional density $p(\mathbf{z} \mid \mathbf{z}_i)$. Since this density is unknown, it is approximated using
 102 a Parzen window estimator based on the available samples $\{\mathbf{z}_1, \dots, \mathbf{z}_{T'}\}$. Under
 103 this approximation, the discrete distribution induced by the kernel is defined
 104 as:

$$P(\mathbf{z}_j \mid \mathbf{z}_i) \approx \frac{k(\mathbf{z}_i, \mathbf{z}_j; \mathbf{W}_Q, \mathbf{W}_K)}{\sum_{j'=1}^{T'} k(\mathbf{z}_i, \mathbf{z}_{j'}; \mathbf{W}_Q, \mathbf{W}_K)} \quad (5)$$

105 where $k : \mathbb{R}^F \times \mathbb{R}^F \rightarrow \mathbb{R}^+$ is a positive kernel function parameterized by the
 106 projection matrices $\mathbf{W}_Q$ and $\mathbf{W}_K$. Under this formulation, the attention mecha-
 107 nism can be interpreted as the expected value of the transformed representations
 108 $\mathbf{W}_V \mathbf{z}_j$ with respect to the kernel-induced distribution:

$$\text{Attention}(\mathbf{Z})_i = \sum_{j=1}^{T'} P(\mathbf{z}_j \mid \mathbf{z}_i) \mathbf{W}_V \mathbf{z}_j = \mathbf{W}_V \mathbb{E}_{P(\mathbf{z}_j \mid \mathbf{z}_i)}[\mathbf{z}_j] \quad (6)$$

109 In this way, kernel-based attention admits a clear probabilistic interpreta-
 110 tion: the output $\text{Attention}(\mathbf{Z})_i$ corresponds to the expected value of the value
 111 representations under a probability distribution induced by the kernel.

112 4 Experimental Setup

113 4.1 Dataset Description

114 The study employs multichannel electroencephalographic (EEG) recordings from
 115 the public dataset “EEG Data for ADHD / Control Children” [4]. The database
 116 contains signals from $S = 121$ subjects, of which 61 belong to the ADHD class
 117 and 60 to the control class. Each subject is assigned a binary label $y_s \in \{0, 1\}$,
 118 where 0 denotes control and 1 denotes ADHD.

119 EEG acquisition follows the international 10–20 system using 19 electrodes
 120 distributed across frontal, central, temporal, parietal, and occipital regions, as
 121 illustrated in Figure 1. The recorded channels include Fp1, Fp2, F3, F4, F7, F8,
 122 Fz, Cz, Pz, C3, C4, T3, T4, T5, T6, P3, P4, O1, and O2, with references placed
 123 on the earlobes (A1, A2). Signals were sampled at a frequency of $f_s = 128$
 124 Hz. Each recording corresponds to a continuous EEG signal acquired during a
 125 visual attention paradigm, in which participants perform a counting task based
 126 on sequentially presented visual stimuli.

127 4.2 Preprocessing

128 The preprocessing pipeline transforms continuous EEG recordings into short-
 129 duration trials suitable for model training. A Butterworth bandpass filter in
 130 the range $[0.5, 60]$ Hz removes low-frequency drifts and high-frequency noise.
 131 Furthermore, a notch filter at 50 Hz suppresses power-line interference.

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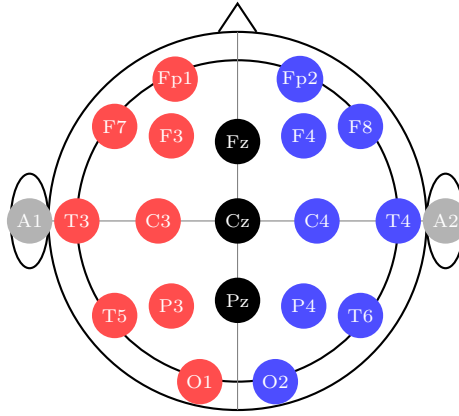


Figure 1: EEG acquisition setup under the 10–20 system.

132 The filtered signal is subsequently segmented using a sliding window of length
 133 $L = 2$ seconds with 50% overlap. This segmentation produces multiple trials
 134 per subject, consistent with the formulation described in Section 3.1. The use
 135 of overlapping windows increases the number of training samples and maintains
 136 temporal continuity between adjacent segments, improving model robustness in
 137 scenarios with limited subject availability and non-stationary EEG dynamics
 138 [5, 6].

139 Figure 2 illustrates the distribution of N_s across subjects, showing variability
 140 in the number of generated trials due to differences in recording duration.

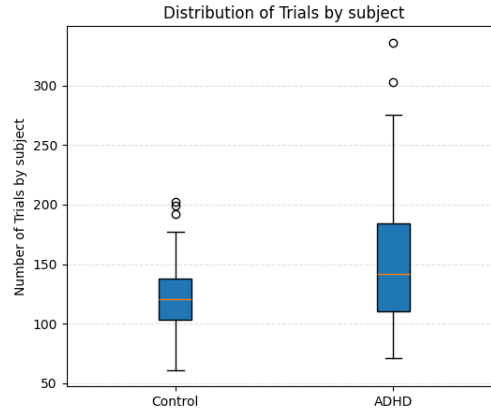


Figure 2: Distribution of trials per subject.

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141 4.3 Nested Group K-Fold Evaluation

142 Model evaluation follows a nested group-based cross-validation scheme. Let $\mathcal{S}$
 143 denote the set of subjects. A Group K -Fold partition with $K = 10$ divides $\mathcal{S}$
 144 into disjoint subsets $\{\mathcal{S}^{(k)}\}_{k=1}^K$. At each outer iteration k , the subset $\mathcal{S}^{(k)}$ defines
 145 the test set, while the remaining subjects form the development set. A second
 146 split is then applied to the development set to separate training and validation
 147 subsets, preserving subject-level grouping.

148 Model parameters are optimised by minimising the binary cross-entropy loss:

$$l_{\text{BCE}}(y_{s,n}, \tilde{y}_{s,n}) = \sum_{n=1}^{N_s} [y_{s,n} \log(\tilde{y}_{s,n}) + (1 - y_{s,n}) \log(1 - \tilde{y}_{s,n})] \quad (7)$$

$$\mathcal{L} = - \sum_{s=1}^S l_{\text{BCE}}(y_{s,n}, \tilde{y}_{s,n}) \quad (8)$$

149 Performance metrics, including accuracy, precision, recall, and F1-score,
 150 were computed from the confusion matrix. Since each subject is associated
 151 with multiple trials and all trials inherit the subject-level label, the evaluation
 152 is conducted at the subject level. To this end, a majority voting scheme is
 153 applied to the model predictions, as shown below:

$$\hat{y}_s = \arg \max_{c \in \{0,1\}} \sum_{n=1}^{N_s} l_{\text{BCE}}(c, \tilde{y}_{s,n}) \quad (9)$$

154 4.4 Model Configuration

155 Hyperparameter selection is performed using Bayesian optimisation over the
 156 validation sets. This process explores configurations that balance model ca-
 157 pacity and generalisation. The model consists of three main components: (i) a
 158 convolutional embedding module, (ii) a transformer encoder with a kernel-based
 159 attention mechanism, and (iii) a classification head.

160 The architecture incorporates non-linear activation functions, layer normal-
 161 isation within transformer blocks, and dropout as a regularisation mechanism.
 162 These components support stable training and reduce overfitting without in-
 163 creasing model complexity. The optimisation process tunes the main hyperpa-
 164 rameters of the model, including: Embedding dimension F' , Number of trans-
 165 former layers L_{tr} , Number of attention heads H , Feed-forward dimension d_{ff} ,
 166 Number of convolutional filters F , Temporal kernel size k_t , Pooling parameters
 167 (k_p, s_p) , Dropout rate p

168 In particular, the attention mechanism replaces the standard softmax nor-
 169 malisation with a kernel-based formulation, enabling more flexible similarity
 170 measures between query and key representations. This formulation is well suited
 171 for EEG signals, which exhibit non-stationary behaviour and nonlinear depen-
 172 dencies. For this purpose, several kernel functions were evaluated, as shown in
 173 Table 1.

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Table 1: Kernel functions

Kernel	Expression
Standard exponential	$k(\mathbf{Q}_i, \mathbf{K}_j) = \exp(\mathbf{Q}_i^\top \mathbf{K}_j)$
RBF	$k(\mathbf{Q}_i, \mathbf{K}_j) = \exp(-\ \mathbf{Q}_i - \mathbf{K}_j\ ^2)$
Periodic	$k(\mathbf{Q}_i, \mathbf{K}_j) = \exp(-2 \sin^2(\ \mathbf{Q}_i - \mathbf{K}_j\ _2))$

174 To provide a structured overview of the proposed architecture, Table 2 sum-
175 marises the main components of the model, including the embedding stage,
176 transformer encoder, and classification head, along with their corresponding
177 output dimensions and key parameters.

Table 2: Proposed EEG Conformer Architecture

Layer	Output Size	Parameters
Input	EEG Segment	(C, T) –
Patch Embedding		
Temporal Conv	(F, C, T')	$F \cdot k_t$
Spatial Conv	$(F, 1, T')$	$F^2 \cdot C$
Projection	$(F', 1, T')$	–
Transformer Encoder ($\times L$ layers)		
Multi-Head Attention	$(F_{\text{emb}}, T_{\text{emb}})$	–
Add & Norm	$(F_{\text{emb}}, T_{\text{emb}})$	–
Feed Forward	$2F_{\text{emb}}d_{ff}$	–
Add & Norm	$(F_{\text{emb}}, T_{\text{emb}})$	–
Classification Head		
Flatten	$(F \cdot T')$	–
FC1	(256)	–
FC2	(32)	–
Output	(K)	classes

178 5 Results and discussion

179 5.1 Quantitative Results

180 Table 3 reports the classification performance of the model under three kernel-
181 based attention formulations, considering the exponential kernel as the baseline.

182 The results indicate that the RBF kernel consistently achieves the best
183 performance across all evaluated metrics. Additionally, this configuration ex-
184 hibits lower variability across runs, suggesting improved stability with respect
185 to weight initialization and the optimization process. This behavior is partic-
186 ularly evident in the F1-score, where the RBF-based model shows a standard
187 deviation of ± 0.10 , compared to ± 0.17 for the exponential kernel.

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The precision–recall trade-off further highlights distinct decision behaviors induced by each kernel. The periodic kernel attains a high recall (0.94 ± 0.11), but at the expense of a substantial drop in precision (0.67 ± 0.22), indicating a higher false positive rate and a tendency toward less restrictive decision boundaries. In contrast, the RBF kernel achieves a more balanced trade-off, with precision and recall values of 0.79 ± 0.12 and 0.95 ± 0.08 , respectively, resulting in the highest overall F1-score.

Overall, these results demonstrate that the choice of similarity function within the attention mechanism has a direct impact on the model’s discriminative capability. In particular, the RBF kernel enables the learning of more robust and stable representations, improving both average performance and consistency across runs.

Table 3: Performance comparison across different attention kernels

Kernel	Accuracy	F1-score	Precision	Recall	AUC
Exponential	0.81 ± 0.14	0.82 ± 0.17	0.75 ± 0.17	0.91 ± 0.19	0.89 ± 0.08
Periodic	0.69 ± 0.23	0.76 ± 0.16	0.67 ± 0.22	0.94 ± 0.11	0.78 ± 0.24
RBF	0.84 ± 0.10	0.86 ± 0.10	0.79 ± 0.12	0.95 ± 0.08	0.90 ± 0.09

5.2 Embedding Space Analysis

To better understand how the proposed model transforms the feature space, a t-SNE projection is computed at different stages of the transformer, including the input embedding, intermediate layers, and the final representation. Figure 3 illustrates the evolution of the embedding space across layers. At the input stage, representations from both classes (Control and ADHD) exhibit a high degree of overlap, indicating that the initial feature space is not linearly separable and lacks a clear discriminative structure.

As the representations propagate through the transformer layers, a progressive organization of the feature space can be observed. In the early layers, partial grouping begins to emerge, although a considerable degree of inter-class mixing remains. At intermediate depths, the embeddings start to form more structured manifolds, where samples belonging to the same class become increasingly localized, suggesting the emergence of latent patterns relevant to the classification task. In deeper layers, this effect becomes more pronounced, with class-specific clusters appearing more distinctly separated. This behavior indicates that the transformer encoder learns hierarchical representations that emphasize discriminative features while suppressing irrelevant variability.

At the output of the model, the embedding space exhibits compact and well-defined clusters, reflecting a high degree of class separability. Nevertheless, a transition region characterized by partial overlap between classes can still be observed. This region can be interpreted as an uncertainty zone, where the model exhibits lower confidence in label assignment. In contrast, samples located farther from this boundary tend to correspond to predictions with higher

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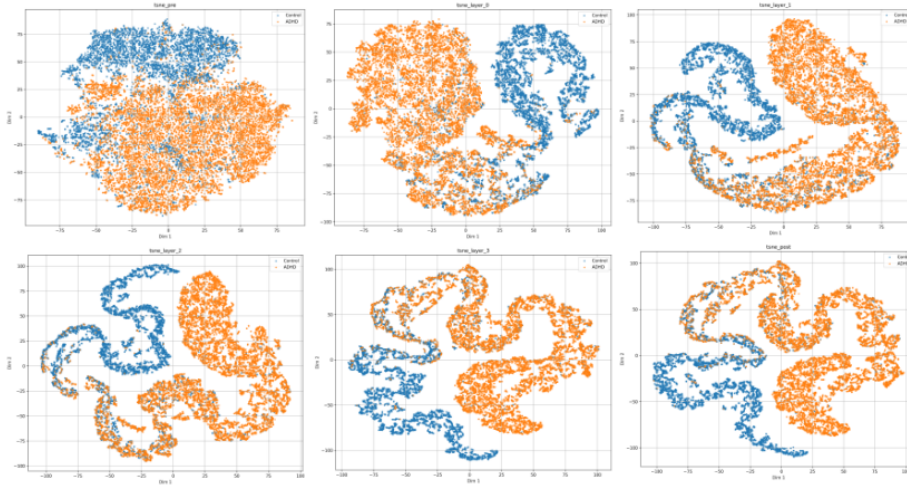


Figure 3: Caption

confidence, suggesting that the geometry of the latent space is aligned with the model’s confidence estimates.

Additionally, a higher dispersion is observed in the representations associated with the ADHD class. This behavior may be attributed to the inherently heterogeneous nature of this population, which is characterized by greater inter-subject variability in neurophysiological patterns. Furthermore, factors such as differences in symptom severity, clinical subtypes, or the presence of noise in EEG recordings may contribute to a more spread-out representation in the latent space. Overall, this behavior supports the hypothesis that the attention mechanism, particularly under the proposed kernel-based formulation, effectively reshapes the feature space into a more discriminative representation. Taken together, these results provide qualitative evidence that the model enhances class separability across layers, transforming an initially entangled representation into a structured latent space suitable for classification.

5.3 Statistical Significance Analysis **Pendiente por redactar**

To assess whether the observed performance differences are statistically significant, a permutation test is conducted between the best-performing baseline and the proposed model. +

Both models are trained multiple times using different random seeds to account for variability in the optimization process.

Figure 4 illustrates the null distribution obtained from the permutation procedure, along with the observed difference in performance.

The resulting p-value is significantly below conventional thresholds (e.g., $p < 0.05$), indicating that the improvement achieved by the proposed model is

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unlikely to be due to random chance.

Additionally, the effect size (Cohen's $d = X$) suggests a moderate to strong practical significance, reinforcing the robustness of the proposed approach.

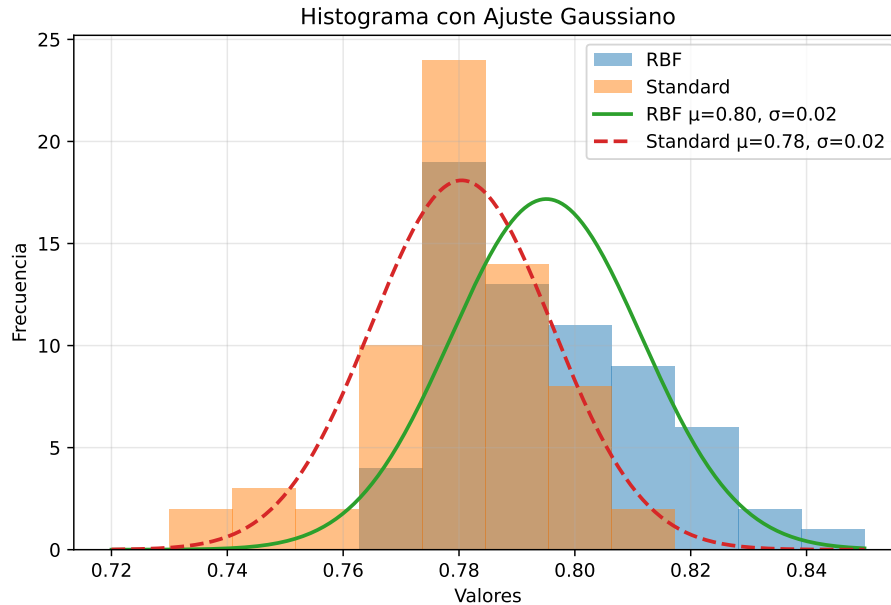


Figure 4: Caption

En la Figure 4 el 0.84 viene del acuacry de mi semilla original- discusion pendiente mis hyperpametros vienen de las desiciones tomadas de esa configuracion

5.4 Future work

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